



Vlaanderen
is supercomputing

Introduction to High Performance Computing

By: VSC HPC Support Staff at KU Leuven site



Training material

Course material is available on Github:
<https://hpcleuven.github.io/HPC-intro/>

Video recordings

- ✓ Recommended videos: ~2 hrs
- ✓ Optional videos: ~1 hrs

Remark

By registering and participating in VSC trainings, you accept to abide by the [VSC training Code of Conduct](#)



What is High Performance Computing?

Concept

Using many computers together to solve large-scale, complex problems faster than one computer can (parallelism).

Applications

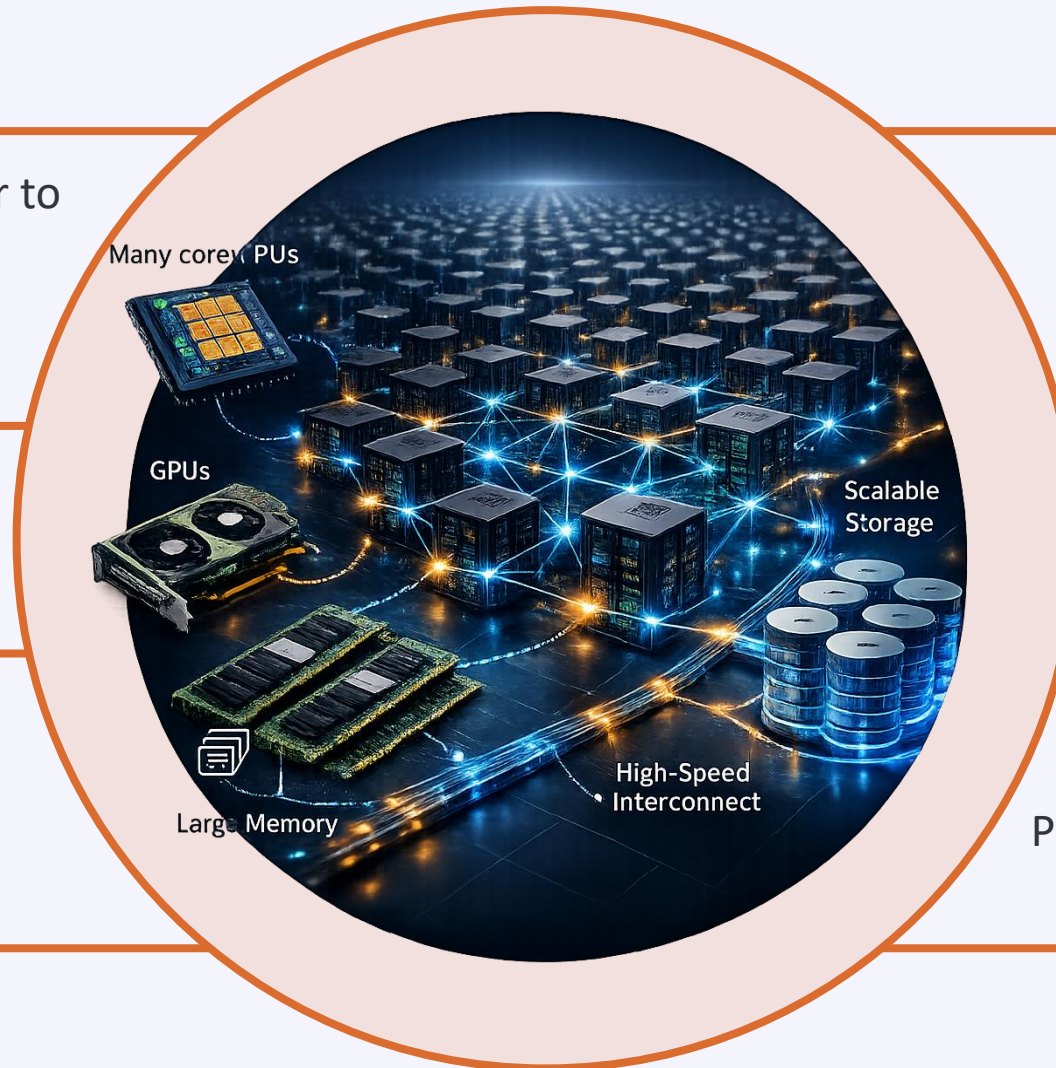
Numerical (CFD) simulations
Molecular dynamics
Long-term weather predictions
Machine learning and AI
Genomics

Hardware Layer

Hundreds of nodes
Tens of cores per node
Large memory
Fast interconnect
GPUs and Network filesystem

Software Layer

Linux OS + system libraries
(GCC) Compilers
Parallel libraries (OpenMP, MPI, ...)
Parallelized software



Training outline

Part 1: Basics

- ✓ About VSC
- ✓ Tier-2 clusters
- ✓ Storage
- ✓ Compute credits
- ✓ Connecting to a login node
- ✓ Data transfer

Part 2: Hands-on

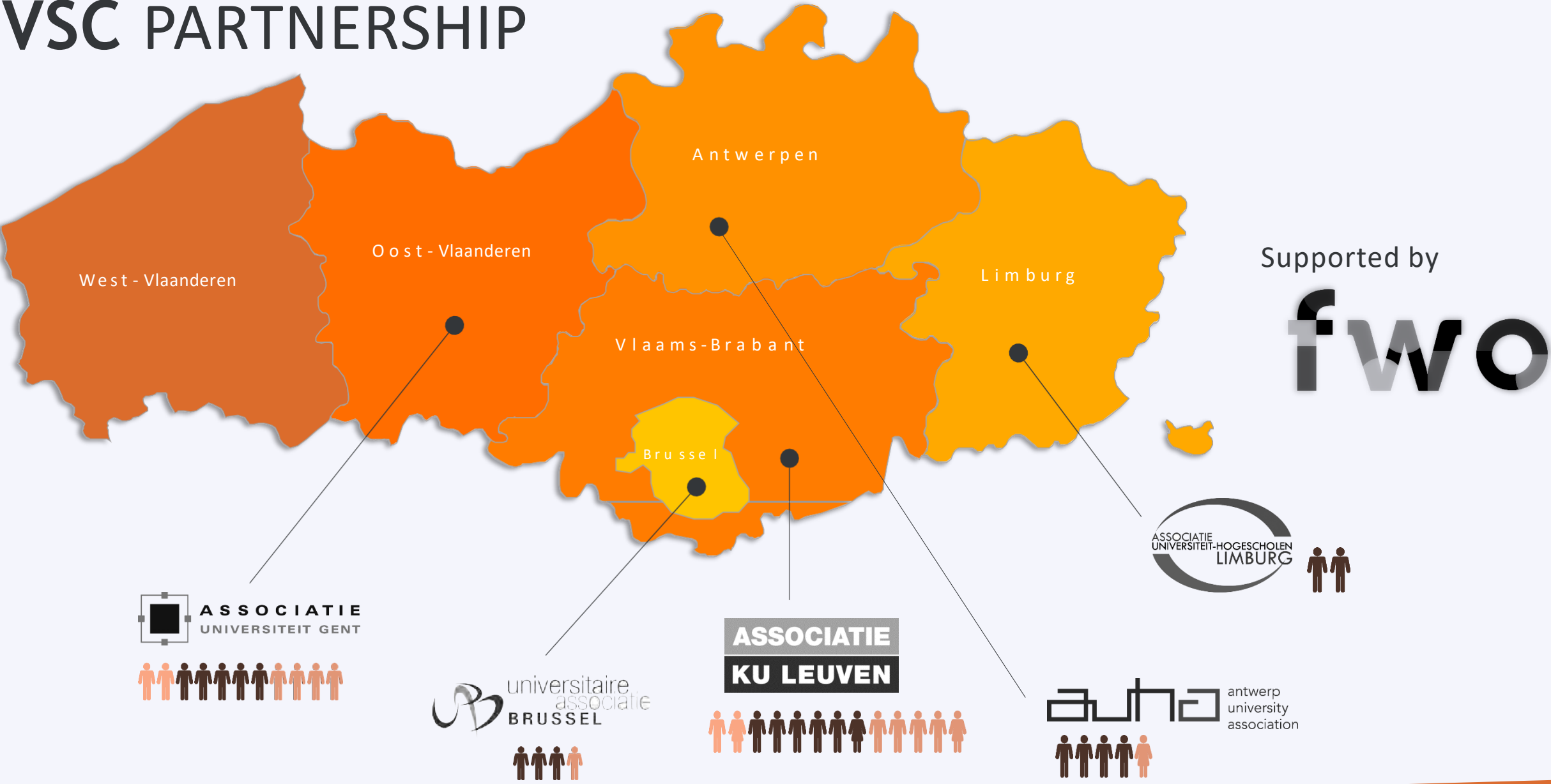
- ✓ Using Open OnDemand
- ✓ Software
- ✓ Starting to compute
- ✓ Demo



VSC

(Vlaams Supercomputer Centrum)

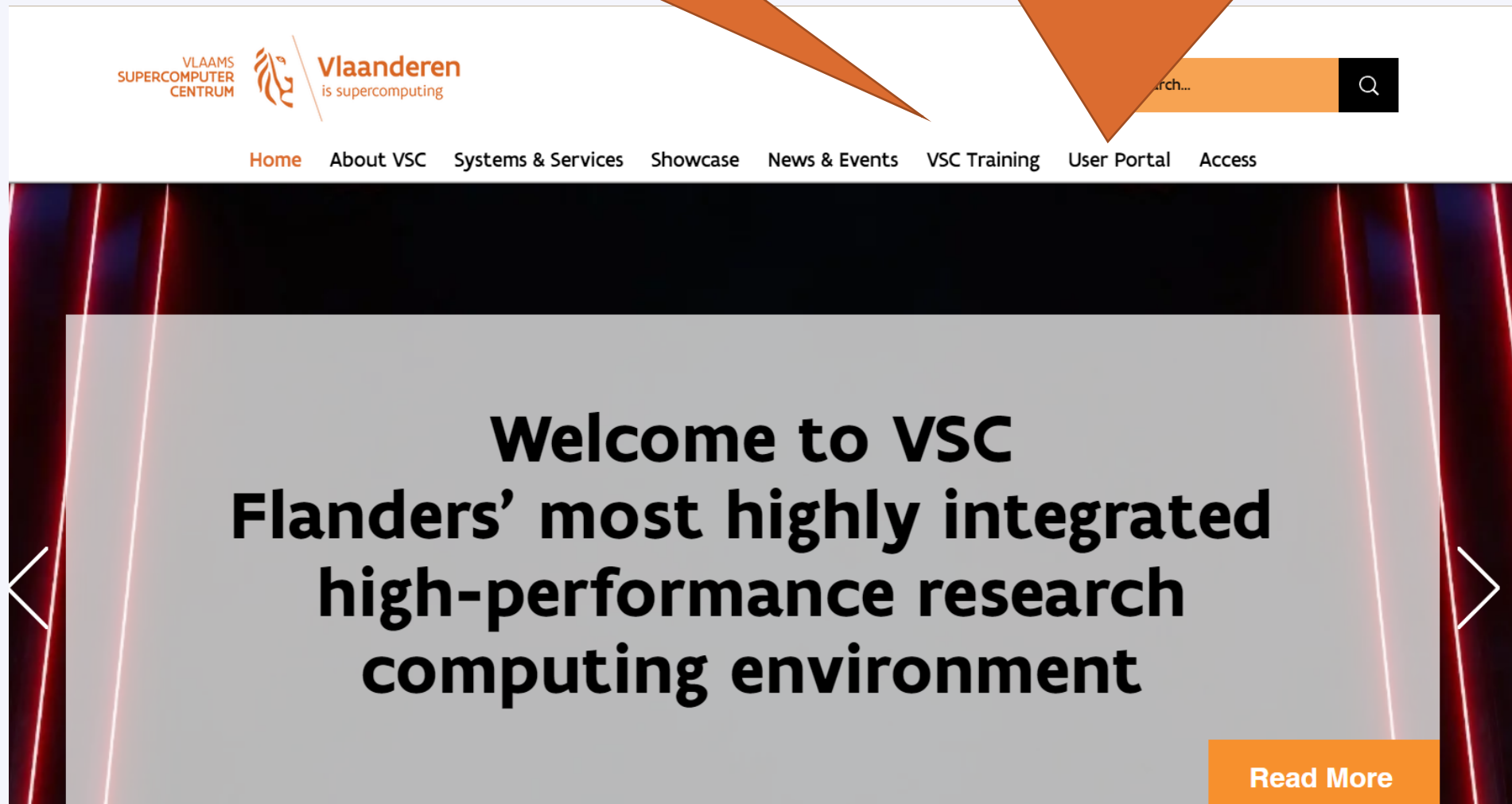
VSC PARTNERSHIP



Visit <https://vscentrum.be>

Extensive set of courses
every academic year

Links to the VSC **account page** and the VSC **documentation**
(answers >80% of your questions!)





Welcome to the User Portal

Here you can find the gateway to the User documentation of the Vlaams Supercomputer Centrum

User documentation page

Manage your VSC-account

partner institute account required

VSC account page

<https://docs.vscentrum.be>

The screenshot shows a web browser window with the URL <https://docs.vscentrum.be>. The page header includes the VLAAMS SUPERCOMPUTER CENTRUM logo and navigation links: About, Contact, Accounts, Compute, Data, Cloud, and FAQs. A search icon is located in the top right corner of the header. An orange callout box with an arrow points to this search icon, containing the text: "Enter your keywords here; e.g. 'OnDemand'". The main content area has a dark background and features a large heading "Welcome to the VSC documentation". Below this, a paragraph states: "The VSC documentation offers extensive *how-to* guides and technical information about the services provided by the Vlaams Supercomputer Centrum." The page is organized into sections: "VSC Accounts" (with a sub-description: "How to get your VSC account to use the different VSC services and platforms."), "Compute" (describing high-performance computing tiers), "Data" (describing research data management), and "Cloud" (describing on-demand resources). Each section has associated sub-sections listed below it: "Tier-1 HPC", "Tier-2 HPC", and "Terminal Interface" under Compute; "Tier-1 Data" under Data; and "Virtual Machines" under Cloud.

Welcome to the VSC documentation

The VSC documentation offers extensive *how-to* guides and technical information about the services provided by the Vlaams Supercomputer Centrum.

VSC Accounts
How to get your VSC account to use the different VSC services and platforms.

Compute
The high-performance computing (HPC) platform provides multiple tiers of parallel processing enabling researchers to run advanced application programs efficiently, reliably and quickly.

Data
The VSC Data component enables research data to remain close to the computing infrastructure during the active phase of the data life cycle.

Cloud
The VSC Cloud component provides *on-demand* resources in a more flexible and cloud-like manner.

Tier-1 HPC Tier-2 HPC Tier-1 Data Virtual Machines

Manage your VSC account at
<https://account.vscentrum.be>

The screenshot shows a web browser window titled "View account — VSC" with the URL account.vscentrum.be/django/ in the address bar. The page features the VLAAMS SUPERCOMPUTER CENTRUM and Vlaanderen is supercomputing logos. A navigation bar contains buttons: View Account, Edit Account, View Groups, New/Join Group, Edit Group, New/Join VO, Reservations, and Log Out. The "View Account" button is highlighted with an orange bar. Below the navigation bar, the "View account" section is active, showing "General information" with fields for Uid, Institute (Leuven / Hasselt), Institute login, and Gecos. Several orange callout boxes with arrows point to specific features: "Add/remove your public keys" points to the "Edit Account" button; "View public keys Request account" points to the "View Account" button; "Create or request to join a new group" points to the "New/Join Group" button; and "Moderate your VSC groups (add/remove users)" points to the "Edit Group" button.

VLAAMS SUPERCOMPUTER CENTRUM Vlaanderen is supercomputing

Add/remove your public keys

View Account Edit Account View Groups New/Join Group Edit Group New/Join VO Reservations Log Out

View public keys Request account

View account

General information

Uid:
Institute: Leuven / Hasselt
Institute login:
Gecos:

Create or request to join a new group

Moderate your VSC groups (add/remove users)

Support and services

Basic support

- Helpdesk (hpcinfo@kuleuven.be)
- Monitoring and reporting

Application support

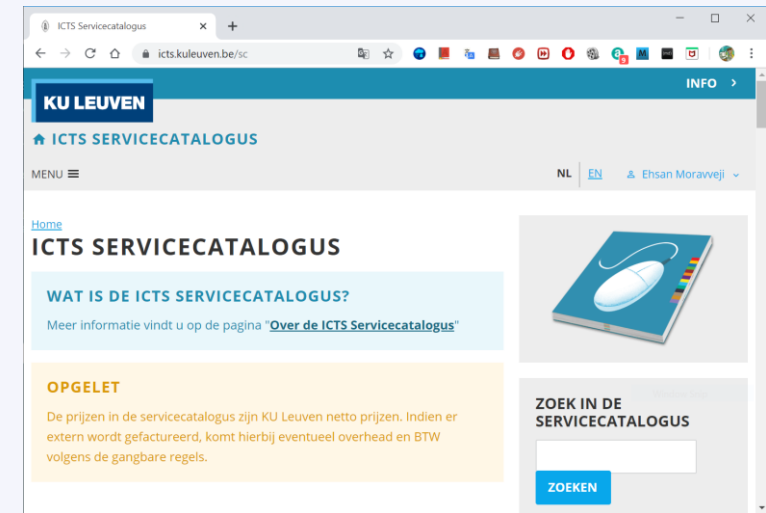
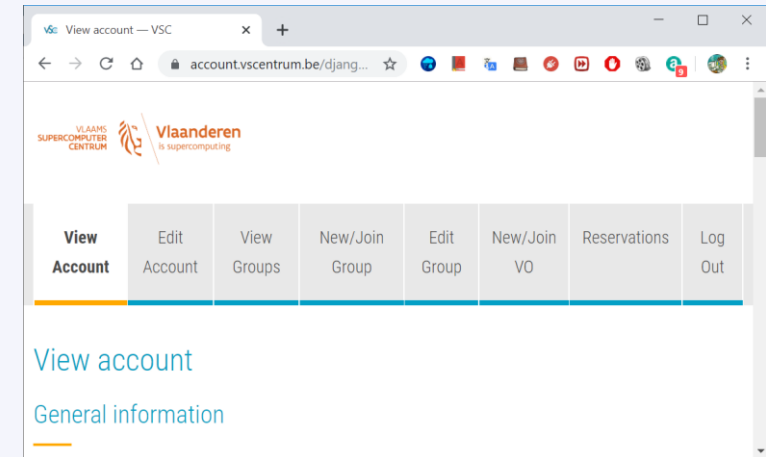
- Installation and porting
- Optimisation and debugging
- Benchmarking
- Workflows and best practices

Training

- Documentation and tutorials
- Scheduled trainings and workshops
- Workshops on request
- One-to-one sessions

Join VSC: become a user

- ✓ Request a [VSC account](#)
- ✓ Request [introductory credits](#) (2M free credits for 6 months)
- ✓ Request [project credits](#) (for supervisors and project leaders)
 - create a [VSC group](#)
 - Add users to the group to give them access to use credits
 - Fill out the [request form](#)
- ✓ Extra [storage](#) requests
 - Scratch extension: free of charge
 - Project storage: 20 € per TB per year
- ✓ All service costs (compute and storage) are described in the ICTS service catalog: <https://icts.kuleuven.be/sc>
Click on [High Performance Computing](#) (NL/EN)



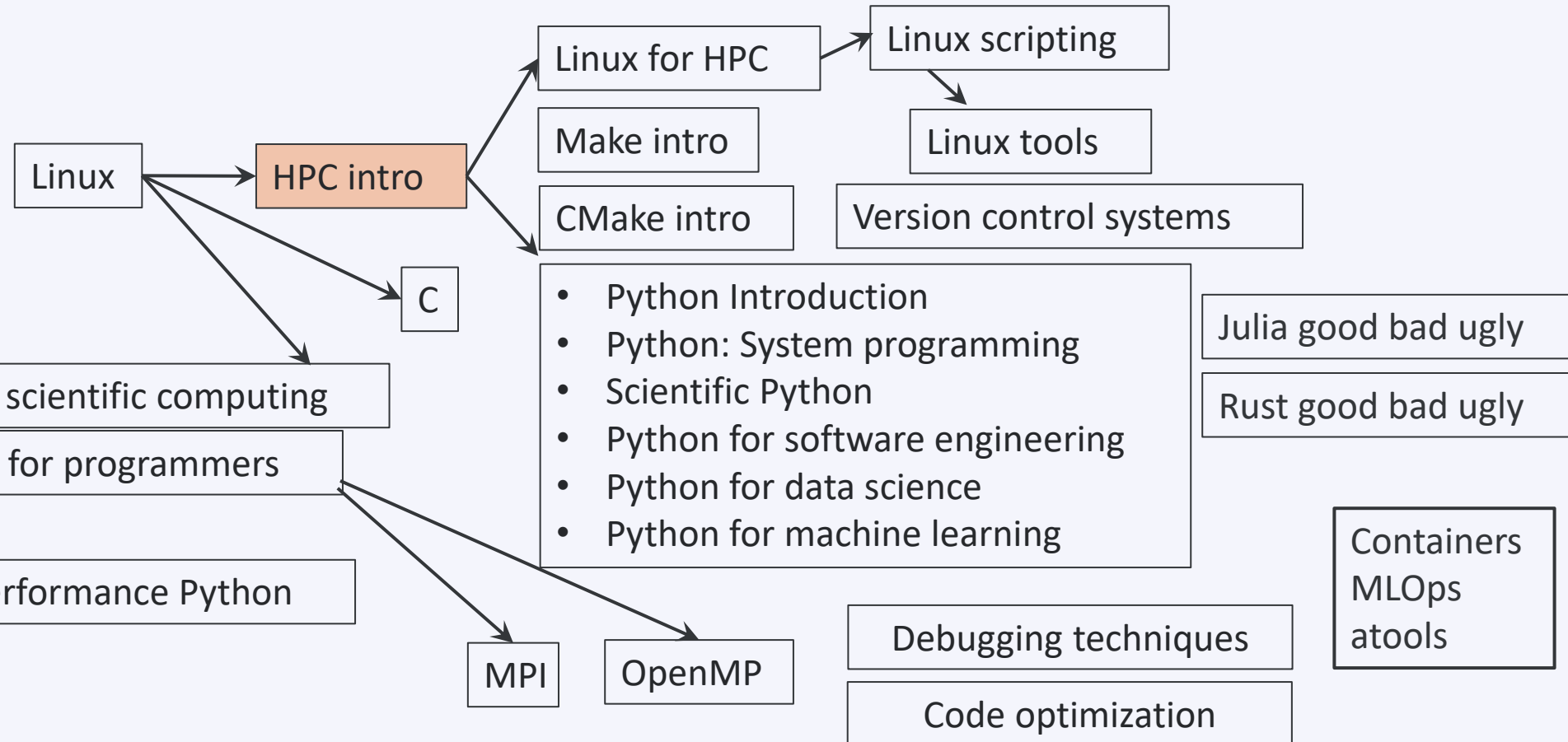
VSC training

- Introductory

- Intermediate

- Advanced

- Specialized track



Stay up-to-date <https://www.vscentrum.be/training>

Overview (for trainings given by Dr. G. J. Bex): <https://gibex.github.io/Training-sessions/>

How to acknowledge VSC in publications

Why?

- ✓ a contractual obligation for the VSC
- ✓ helps VSC to secure funding
- ✓ you will benefit from it in the long run

For KU Leuven users

- ✓ add the relevant papers to the “KUL-HPC” virtual collection in Lirias

In het Nederlands

De infrastructuur en dienstverlening gebruikt in dit werk werd voorzien door het VSC (Vlaams Supercomputer Centrum), gefinancierd door het FWO en de Vlaamse overheid.

In English

The resources and services used in this work were provided by the VSC (Flemish Supercomputer Center), funded by the Research Foundation - Flanders (FWO) and the Flemish Government.



Tier-2 clusters

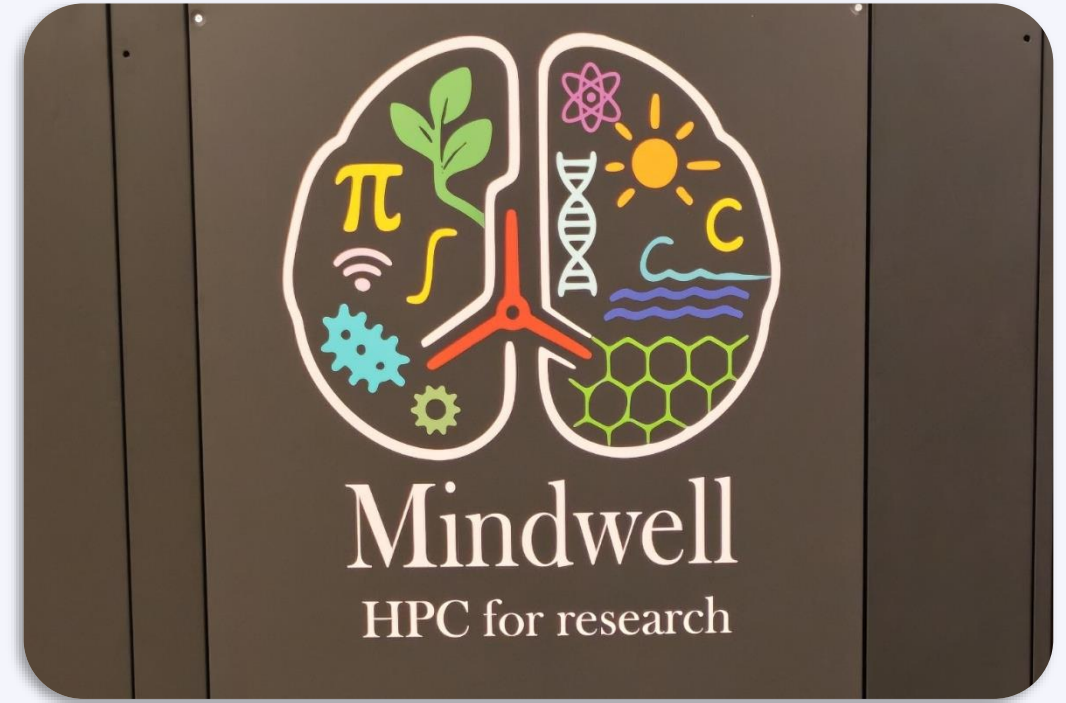
VSC HPC infrastructure



Tier-2 clusters @ KU Leuven



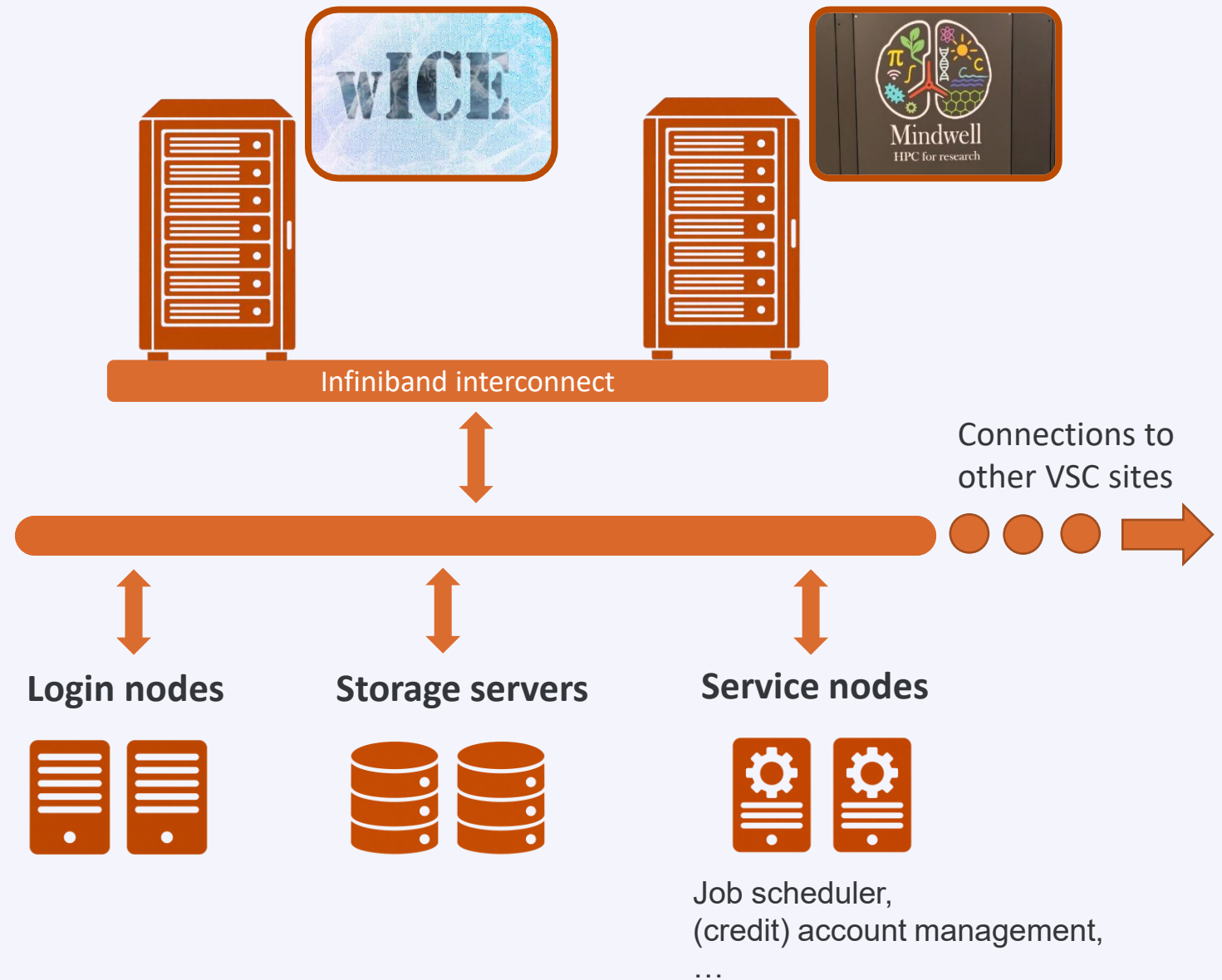
wICE (since September 2022)
276 nodes | 20848 cores



Mindwell (since May 2026)
55 nodes | 10560 cores

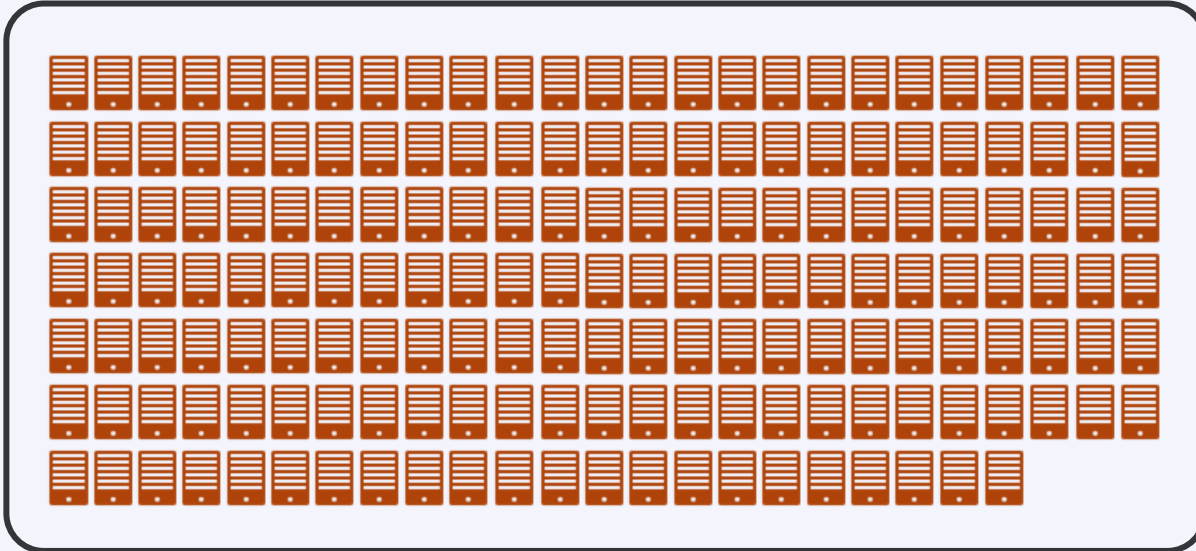
Tier-2 cluster layout

- ✓ Login nodes
- ✓ Storage servers
- ✓ Service nodes
- ✓ Compute nodes
- ✓ Interconnect network



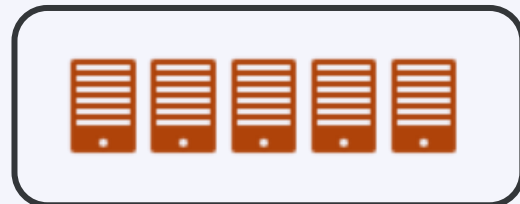
Tier-2 cluster – wICE CPU

Icelake Thin nodes



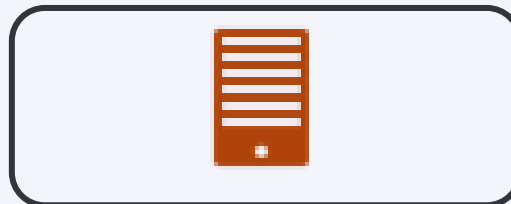
172x IceLake 72c 256 GiB

Large memory nodes



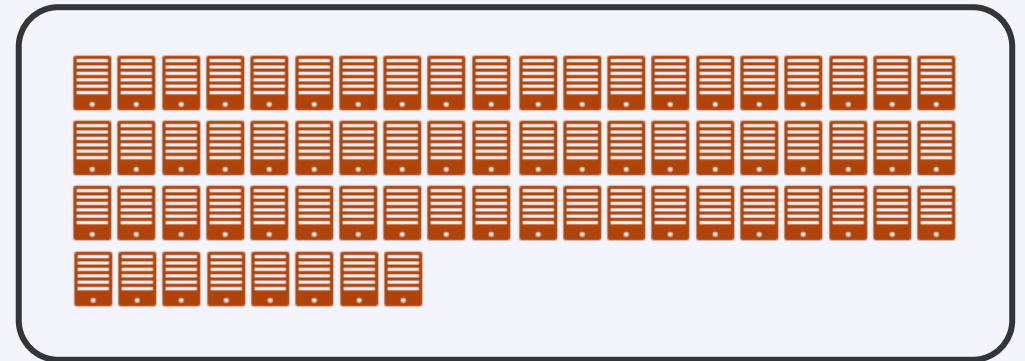
5x IceLake 72c 2 TiB

Huge memory node



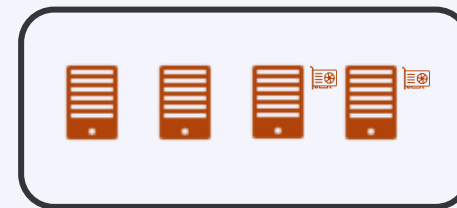
1x IceLake 72c 8 TiB

Sapphire Rapids Thin nodes



68x Sapphire Rapids 96c 256 GiB

Login nodes



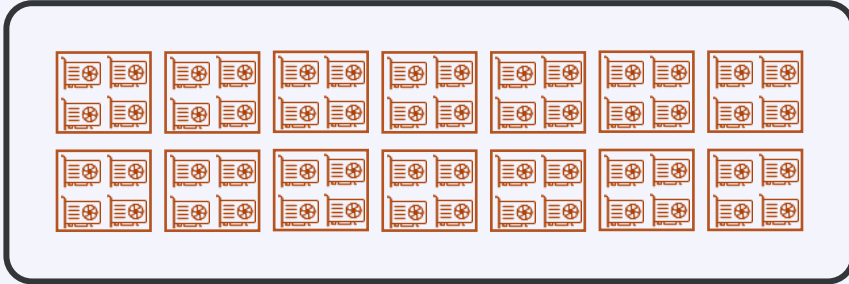
2x SkyLake 36c 377 GiB

2x SkyLake 36c 377 GiB + P6000 24 GiB

Tier-2 cluster – wICE GPU

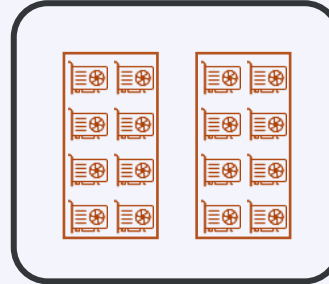


P100 GPU nodes



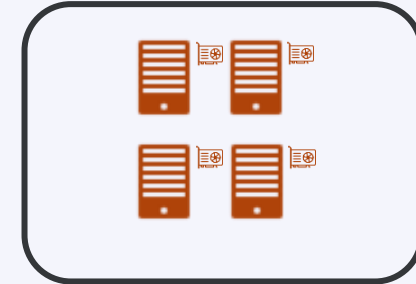
14x SkyLake 36c 192 GiB
4 P100 SXM2 16 GiB

V100 GPU nodes



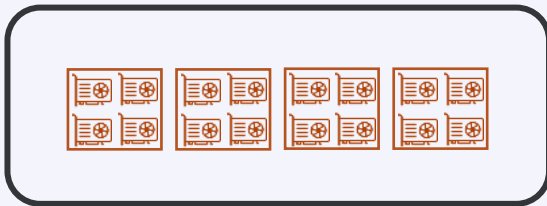
2x CascadeLake 36c 768 GiB
8 V100 SXM2 32 GiB

Interactive nodes



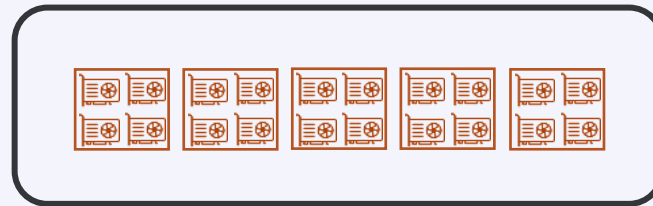
4x IceLake 64c 512 GiB
1 A100 80GB

A100 GPU nodes



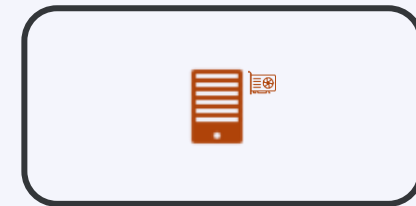
4x IceLake 72c 512 GiB
4 A100 SXM4 80 GiB

H100 GPU nodes



5x AMD Genoa 64c 768 GiB
4 H100 SXM5 80 GiB

A100 Debug node

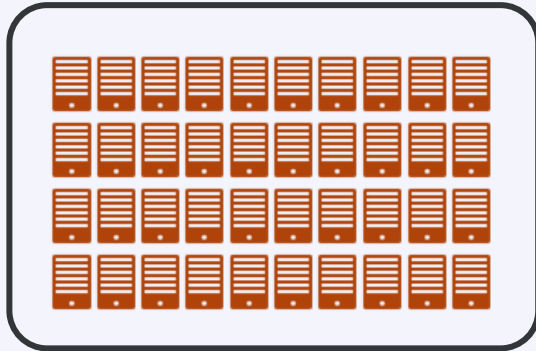


1x IceLake 64c 512 GiB
1 A100 80GB

Tier-2 cluster – Mindwell

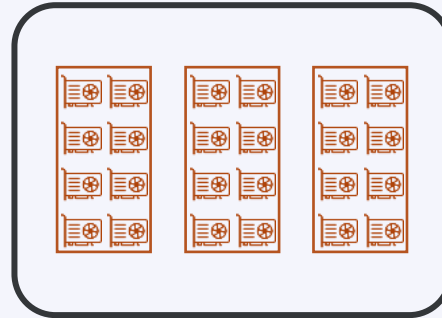


Granite Rapids Thin nodes



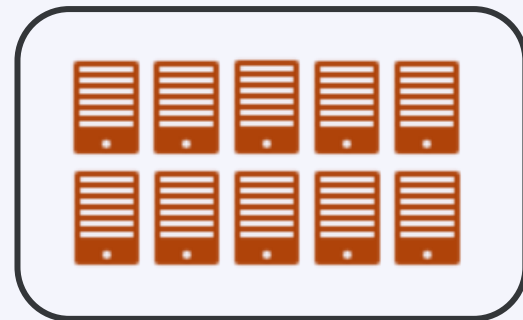
40x Granite Rapids 192c 768 GiB

B200 GPU nodes



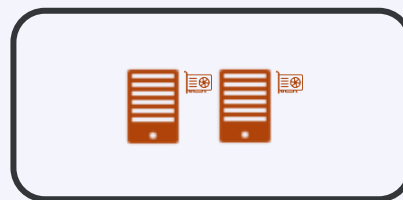
3x AMD Turin 192c 1.5 TiB
8 B200 SXM6 192 GiB

Large memory nodes



10x Granite Rapids 192c 1.5 TiB

Interactive nodes



2x Granite Rapids 192c 768 GiB
2 RTX5000 32 GiB

Login nodes





Tier-2 cluster – wICE CPU

Node type	CPU type	# cores	Total mem	Mem per core (MiB)	Partition
Icelake	Xeon 8360Y	72	256 GiB	3400	batch_icelake
Sapphire Rapids	Xeon 8468	96	256 GiB	2500	batch_sapphirerapids
Icelake large mem	Xeon 8360Y	72	2 TiB	28000	bigmem
Icelake huge mem	Xeon 8360Y	72	8 TiB	111900	hugemem

• OS: Rocky 9.8 • Access: login.hpc.kuleuven.be • Interconnect: InfiniBand HDR (100 Gb/s)	• SSD Disks: 960 GB	Remarks
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Tier-2 cluster – wICE GPU

Node type	CPU & GPU type	# cores	Total mem	Mem per core (MiB)	Partition
Skylake P100 GPU	Xeon Gold 6140 4xP100 SXM2 16GB	36	192 GiB	5000	gpu_p100
Cascadelake V100 GPU	Xeon Gold 6240 8xV100 SXM2 32GB	36	768 GiB	21000	gpu_v100
Icelake A100 GPU	Xeon 8360Y 4xA100 SXM4 80GB	72	512 GiB	7000	gpu_a100
Genoa H100 GPU	Epyc 9334 4xH100 SXM5 80GB	64	768 GiB	11700	gpu_h100
Icelake interactive	Xeon 8358 1xA100 SXM4 80GB	64	512 GiB	2000	Interactive
Icelake A100 debug	Xeon 8358 1xA100 SXM4 80GB	64	512 GiB	7500	gpu_a100_debug

- **OS:** Rocky 9.8
- **Access:** login.hpc.kuleuven.be
- **Interconnect:** InfiniBand HDR (100 Gb/s)

- **SSD Disks:** 960 GB
- **Interactive partition:**
max resources: 8 cores, 1 GPU, 16hr

Remarks

Tier-2 cluster – Mindwell



Node type	CPU type	# cores	Total mem	Mem per core (MiB)	Partition
Granite Rapids	Xeon 6972P	192	768 GiB	4000	batch_graniterapids
Icelake large mem	Xeon 6972P	192	1.5 TiB	8000	bigmem
Icelake A100 GPU	EPYC 9655 8xB200 SXM6 192GiB	192	1.5 TiB	8000	gpu_b100
Icelake interactive	Xeon 6972P 2x RTX5000 32GiB	192	768 GiB	4000	Interactive

- **OS:** Rocky 9.8
- **Access:** login.hpc.kuleuven.be
- **Interconnect:** InfiniBand NDR (400 Gb/s)

- **SSD Disks:** 960 GB
- **Interactive partition:**
max resources: 8 cores, 1 GPU, 16hr

Remarks



Storage

Overview of the Tier-2 storage locations

- ✓ Only you own your files (POSIX)
Users can share folders via [VSC groups](#)
- ✓ A VSC account has 3 default storages (free of charge)
 - \$VSC_HOME
 - \$VSC_DATA
 - \$VSC_SCRATCH
- ✓ You can additionally [request project storage](#)
- ✓ Different storage volumes have different:
 - mount point
 - size and performance
 - use case
 - backup and maintenance policy
- ✓ For more info, see the [ICTS Service Catalog](#) (EN/NL)

Storage locations

- ✓ **Avoid using `/tmp` (`$TMPDIR`) on the login nodes**
It is only 10 GB and is needed by the operating system
Your application can crash if using `/tmp`
- ✓ You are automatically logged into your home folder upon login. Best to immediately switch to some other storage, e.g. `cd $VSC_DATA`
- ✓ Always check your storage balance using `myquota` command

Advanced Tip

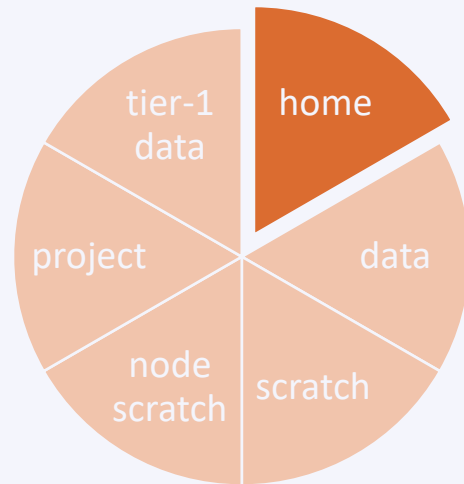
Consider using `/dev/shm` for file operations in memory in your compute jobs

Example

```
$ myquota
file system $VSC_HOME
  Blocks: 1479M of 3072M
  Files: 12934 of 100k
file system $VSC_DATA
  Blocks: 12G of 75G
  Files: 1043k of 10000k
file system $VSC_SCRATCH
  Blocks: 15M of 1.5T
```

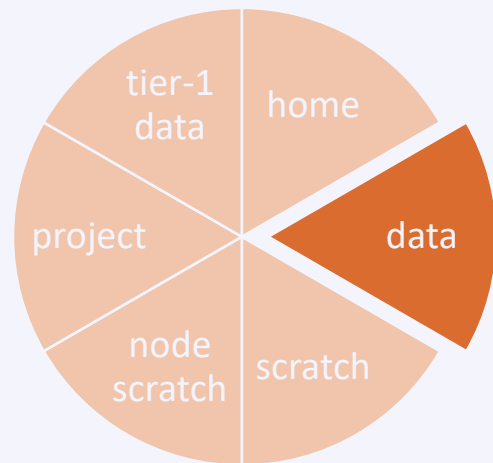
- ✓ [Request form for extra storage](#)
- ✓ [More info in the service catalog](#)

Storage locations



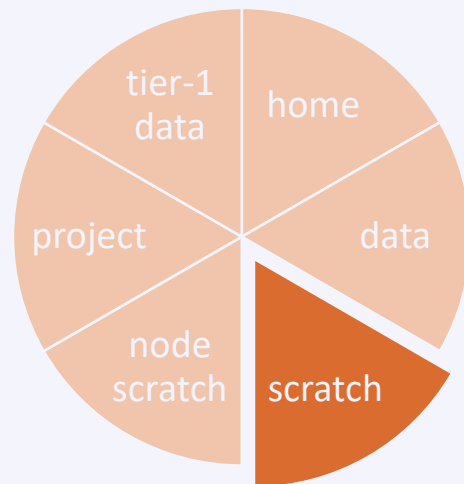
Storage	Home folder
Env. variable	<code>\$VSC_HOME</code>
Filesystem type	NFS
Access	Global
Data Protection	Hourly, daily, weekly (until last month) inside <code>.snapshot</code> folders
Default quota	3 GB
Quota increase	Not possible
Usage	Only storing SSH keys, config files, ...
Remarks	- Stay away from using it - Can easily overflow: + Your jobs may crash + Login issues

Storage locations



Storage	Data folder
Env. variable	\$VSC_DATA
Filesystem type	NFS
Access	Global
Data Protection	Hourly, daily, weekly (until last month) inside <code>.snapshot</code> folders
Default quota	75 GB
Quota increase	On purchase
Usage	Your data, code, software, results
Remarks	- Permanent storage for initial/final results - Not optimal for intensive or parallel I/O

Storage locations

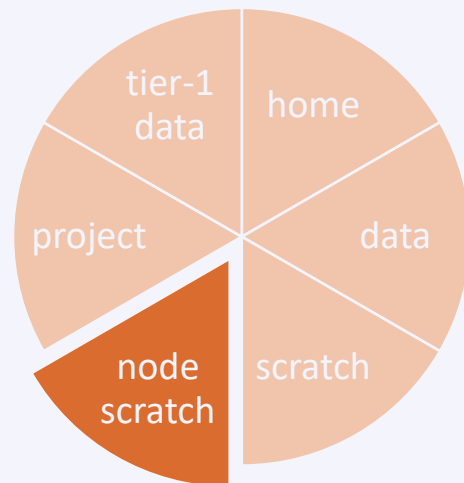


The Lustre filesystem is nearly full.
Momentarily, we don't extend user quota's
with more than 1TB.



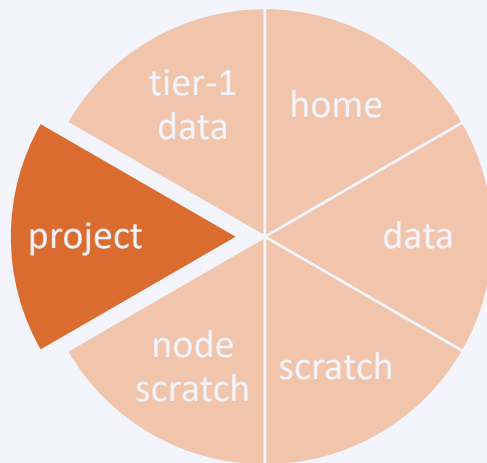
Storage	Scratch folder
Env. variable	<code>\$VSC_SCRATCH</code> <code>\$VSC_SCRATCH_LUSTRE1</code> <code>\$VSC_SCRATCH_GPFS1</code>
Filesystem type	Lustre & GPFS
Access	Local
Data Protection	Files get deleted 30 days after last access
Default quota	500 GB for both
Quota increase	Free
Usage	Intensive, parallel I/O, temporary files
Remarks	- Lustre for wICE, GPFS for Mindwell - Copy scratch files to VSC_DATA or local storage after jobs are done - Deleted files cannot be recovered - Avoid folders with >10,000 files - More info on VSCdocs

Storage locations



Storage	Node scratch folder
Env. variable	<code>\$VSC_SCRATCH_NODE</code>
Filesystem type	Ext4
Access	On compute node, only at runtime
Data Protection	None
Size	Architecture-dependent
Usage	Temporary storage at runtime
Remarks	- Fastest I/O, attached to the node - Is cleaned after job terminates - Copy the data to your home, scratch, or project storage before job ends

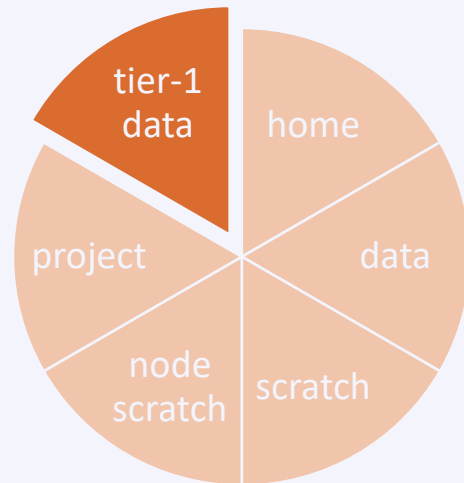
Storage locations



The Lustre filesystem is nearly full.
Momentarily, we don't create or extend
users' staging quota requests.

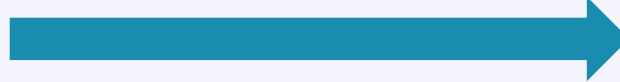
Storage	project folder
Path	/lustre1/project/stg_000XX /gpfs1/project/prj_000XX
Filesystem type	Lustre & GPFS
Access	Local
Data Protection	None
Quota	From 1 TB
Request	- To request, see ICTS Service Catalog - More info on VSC docs
Usage	Permanent; share with a group
Remarks	- Accessible from login/compute nodes - Fast, parallel I/O - Lustre for wICE, GPFS for Mindwell

Storage locations



Storage	project folder
Purpose	Large storage service for active research data
Access	Stage-in/stage-out
Data Protection	snapshots
Quota	From 5 TB
Request	- Free of charge for academic use - Proposal submission - To request, see DATA Vlaams Supercomputer Centrum
Remarks	- Supports metadata

Extra (KU Leuven) storage locations



- ✓ For active data
- ✓ Data should stage in/out
- ✓ Has backup

- ✓ For **inactive** data
- ✓ Storage until 10 years
- ✓ Has backup

Also see the [KU Leuven storage guide](#) for even more storage possibilities

ManGO

Central, secure,
highly available



STORE



Apply rich metadata,
enforce conventions,
standardize

DESCRIBE

AUTOMATE

Implement rule-based
policies & workflows



SHARE



Collaborate securely
& efficiently





Compute credits

Tier-2 credit system

What?

- ✓ Credits are a measure for compute resources
- ✓ Mandatory for accessing these resources
- ✓ Traceability and accountability

Why?

- ✓ Incentive to optimize your code and workflow

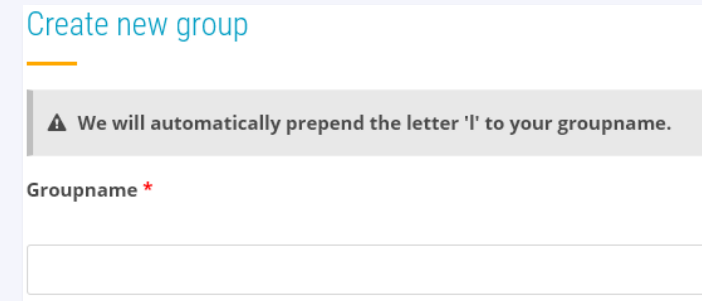
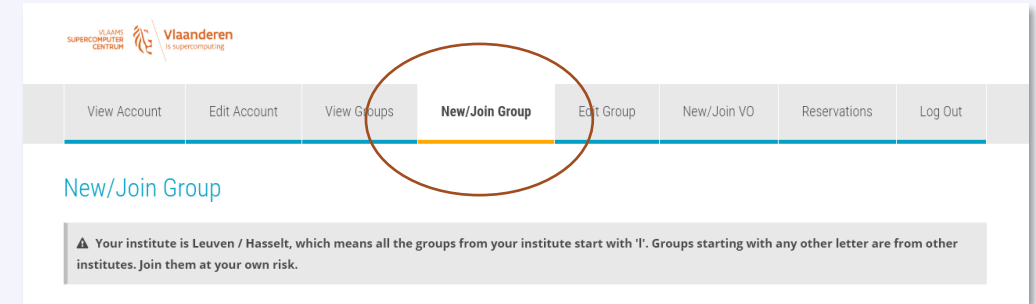
How?

- ✓ 1 million credits = 3.5 EUR
- ✓ Pay-as-you-go model
- ✓ Credit rates on the right = for fully allocated nodes (if you request less cores, you also get billed less)
- ✓ See the [ICTS service catalog](#) for more info

Cluster	Node types	Credits per node-hour
wICE	IceLake Thin	11 000
wICE	Sapphire Rapids Thin	20 000
wICE	Large Memory	19 000
wICE	Hugemem	80 000
wICE	GPU P100 – 4 GPUs	20 000
wICE	GPU V100 – 8 GPUs	39 900
wICE	GPU A100 – 4 GPUs	45 000
wICE	GPU H100 – 4 GPUs	150 000
wICE	Interactive	Free
Mindwell	Granite Rapids Thin	30 000
Mindwell	Large Memory	35 000
Mindwell	GPU B200 – 8 GPUs	245 000
Mindwell	Interactive	Free

Tier-2 credit system – how to request credits

- ✓ Request your introduction credits
 - Every new user can request them once
 - Free of charge
 - 6 months
- ✓ Request permission to join a group
- ✓ Request project credits
 - Create group on the VSC account page
 - Start the name with **p_**
 - Creator of the group can add and remove users
- ✓ Compute credits can be used on all KU Leuven Tier-2 clusters

A screenshot of the 'Create new group' form. The form has a title 'Create new group' in blue. Below the title, there is a warning message: '⚠ We will automatically prepend the letter 'l' to your groupname.' Below the warning, there is a label 'Groupname *' and an empty text input field.

Tier-2 credit system – how

- ✓ Submit your jobs with a suitable credit account

```
$ sbatch --account=lp_myproject myjob.slurm
```

- ✓ Slurm checks if the account has sufficient credits available to complete your job
- ✓ Once a job start running, it starts consuming credits from the account

Tier-2 credit system – manage your credits

- ✓ We provide various `sam`-commands to facilitate credit related queries

Command	Purpose
<code>sam-balance [-A account]</code>	List active projects and available credits

```
tier2-p-login-3$ sam-balance -A lp_hpcinfo
ID          Name          Balance      Reserved      Available
=====
96141      lp_hpcinfo      9782636      0             9782636
tier2-p-login-3$
```

And if you wonder, ‘sam’ stands for Slurm Account Manager.

Tier-2 credit system – manage your credits

- ✓ We provide various `sam`-commands to facilitate credit related queries

Command	Purpose
<code>sam-list-allocations -A <account-name></code>	List the credit deposits and refunds (if any)

```
tier2-p-login-3$ sam-list-allocations -A lp_hpcinfo
AllocID  AccountID Account          Timestamp          Credits
=====  =====  =====
80745    96141     lp_hpcinfo      2023-02-14T10:40:50 2000000
203565   96141     lp_hpcinfo      2023-04-17T16:42:26 2000000
227730   96141     lp_hpcinfo      2023-05-03T09:10:29 2000000
313474   96141     lp_hpcinfo      2023-06-19T17:33:59 8410000
tier2-p-login-3$
```

Tier-2 credit system – manage your credits

- ✓ We provide various `sam`-commands to facilitate credit related queries

Command	Purpose
<code>sam-quote sbatch [sbatch args] [myjob.slurm]</code>	Estimates the (max) amount of credits a job will use

```
tier2-p-login-3$ sam-quote sbatch --cluster=wice hello.slurm
10980
tier2-p-login-3$
```

Tier-2 credit system – manage your credits

- ✓ We provide various `sam`-commands to facilitate credit related queries

Command	Purpose
<code>sam-statement -A <account-name></code> <code>-s <start-date></code> <code>-e <end-date></code>	Overview of credits used by each job in some time window

```
tier2-p-login-3$ sam-statement -A lp_hpcinfo -s 2023-09-01 -e 2023-10-25
#####
#
# Includes Account=lp_hpcinfo
# Generated on Mon Oct 30 11:01:29 2023.
# Reporting fund activity from 2023-09-01 to 2023-10-25
#
#####

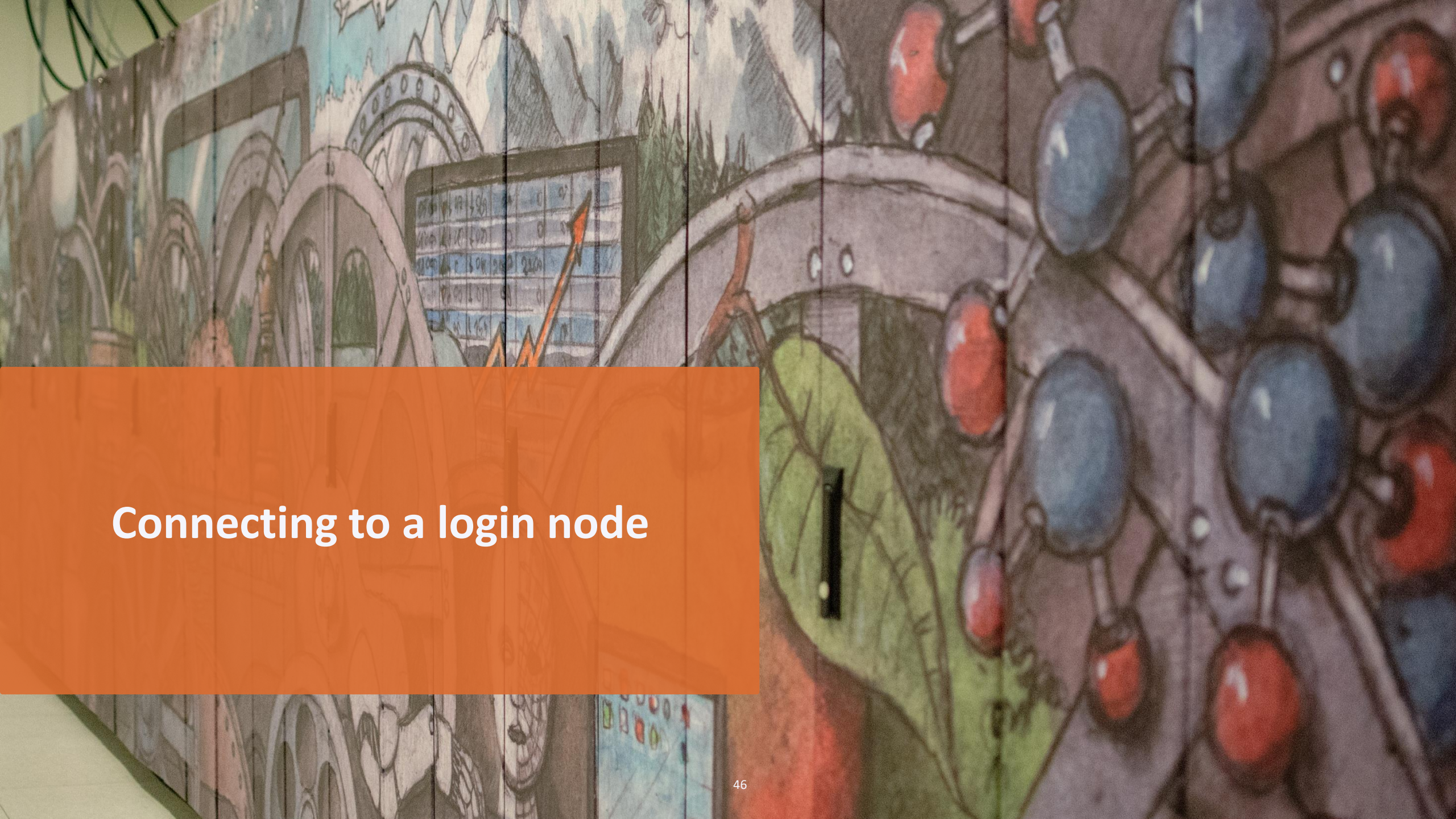
-----
Credits deposited in the given period:          0
Credits refunded in the given period:           0
Credits consumed in the given period:           0
-----

JobID      Cluster  Account      User      Partition  Credits
=====
60739746   wice    lp_hpcinfo   vsc30446  batch      0
60740123   wice    lp_hpcinfo   vsc30446  batch      0
60740125   wice    lp_hpcinfo   vsc30446  batch      0
60746730   wice    lp_hpcinfo   vsc30446  interactive 0
60773547   wice    lp_hpcinfo   vsc30446  interactive 0
60774141   wice    lp_hpcinfo   vsc30446  batch      0
```

Tier-2 credit system – manage your credits

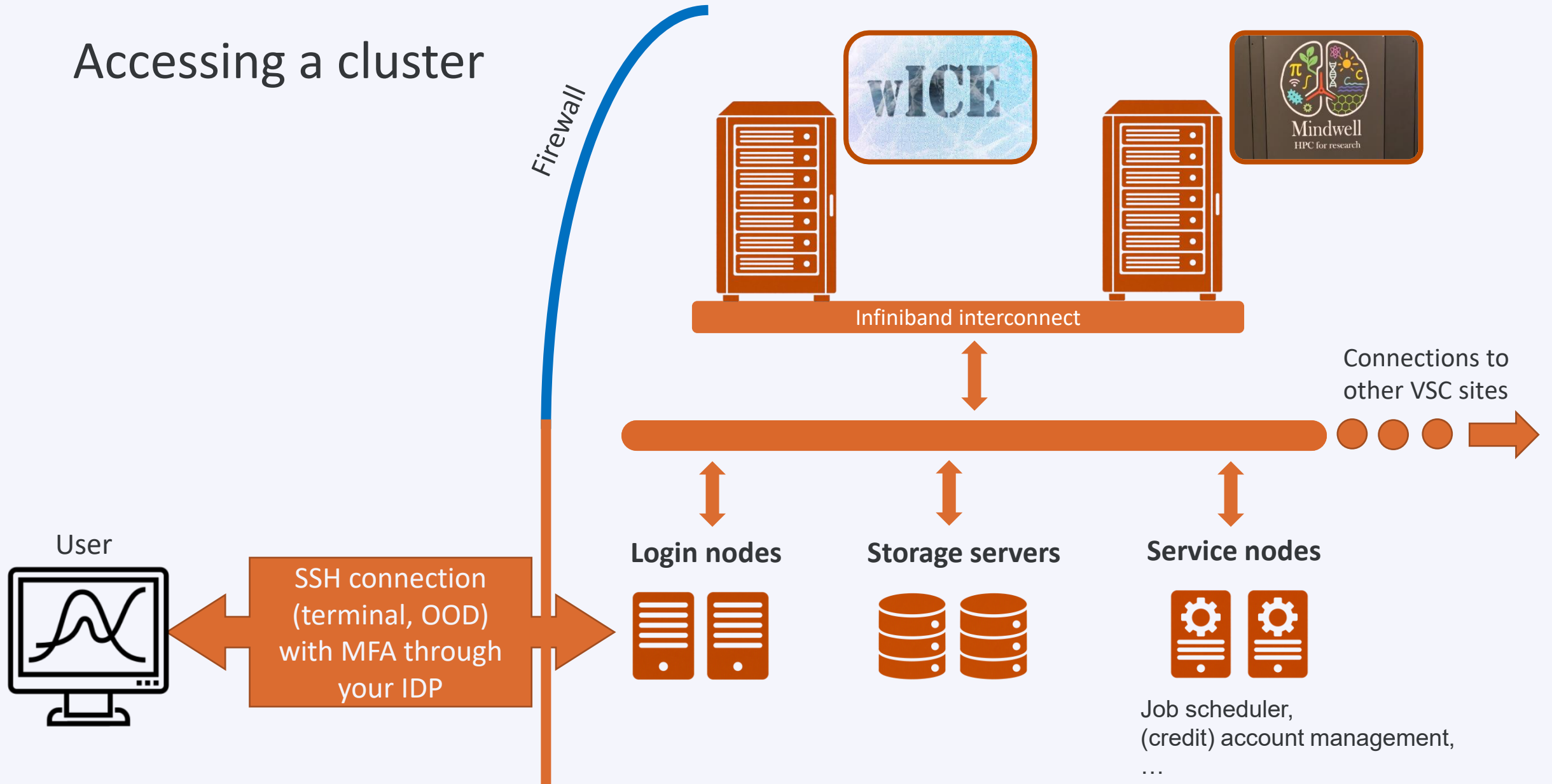
- ✓ We provide various `sam`-commands to facilitate credit related queries

Command	Purpose
<code>sam-balance</code>	Lists active projects and available credits
<code>sam-list-allocations -A <account-name></code>	Lists the credit deposits and refunds (if any)
<code>sam-quote sbatch [sbatch args] [myjob.slurm]</code>	Estimates the (max) number of credits a job will use
<code>sam-statement -A <account-name> -s <start-date> -e <end-date></code>	Overview of credits used by each job in some time window



Connecting to a login node

Accessing a cluster



How to login?

Open OnDemand

- ✓ Web-based login via your browser
- ✓ <https://ondemand.hpc.kuleuven.be>
- ✓ Multi-Factor Authentication (MFA):
Login through your institute's IDP
- ✓ No SSH key needed

SSH client

- ✓ Windows: [PuTTY](#), [MobaXterm](#), ...
MacOS/Linux: any terminal
- ✓ Hosts: login.hpc.kuleuven.be:22
Recommended to use an [SSH agent](#)
- ✓ Can open GUIs if X11 forwarding is enabled
- ✓ [Optional] for VSC clusters not hosted at KU Leuven you need an [\(open\)SSH key pair](#)
(upload the public key to the VSC account page and wait ≥ 30 mins)

Using the login nodes

- ✓ To develop code
- ✓ To check your storage and credit balance
- ✓ To manage jobs (submit, check status, debug, resubmit, ...)
- ✓ To move data around
 - within VSC: data, scratch, project
 - outside VSC: copy/sync from/to your local storage (e.g. Globus)
- ✓ To pre- or postprocess your data
- ✓ To visualize your data
- ✓ To share files and folders



Do NOT execute CPU- or memory-intensive tasks here

Login nodes are shared resources. Submit jobs instead!

Use `slurmtop` to see how busy the cluster is.

Connecting via a terminal

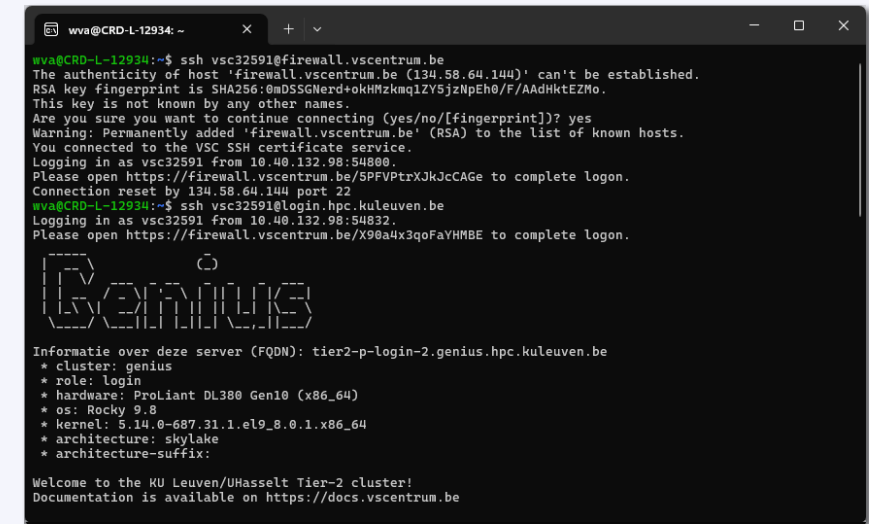
(Windows Powershell, WSL, Linux and Mac)

- ✓ Request/refresh your SSH certificate:

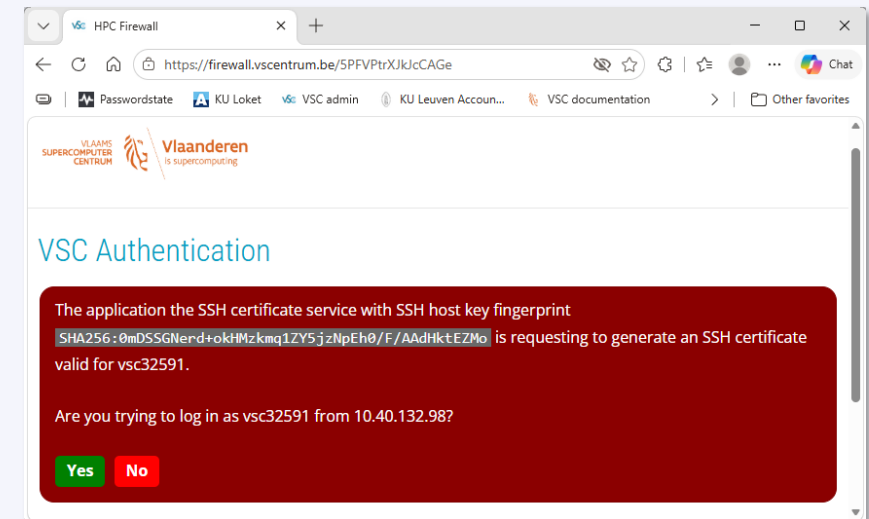
```
$ ssh -A vscXXXXXX@firewall.vscentrum.be
```

- ✓ Use ssh to connect:

```
$ ssh -AX vscXXXXXX@login.hpc.kuleuven.be  
# -A: for agent key forwarding  
# -X: to enable X11 forwarding
```



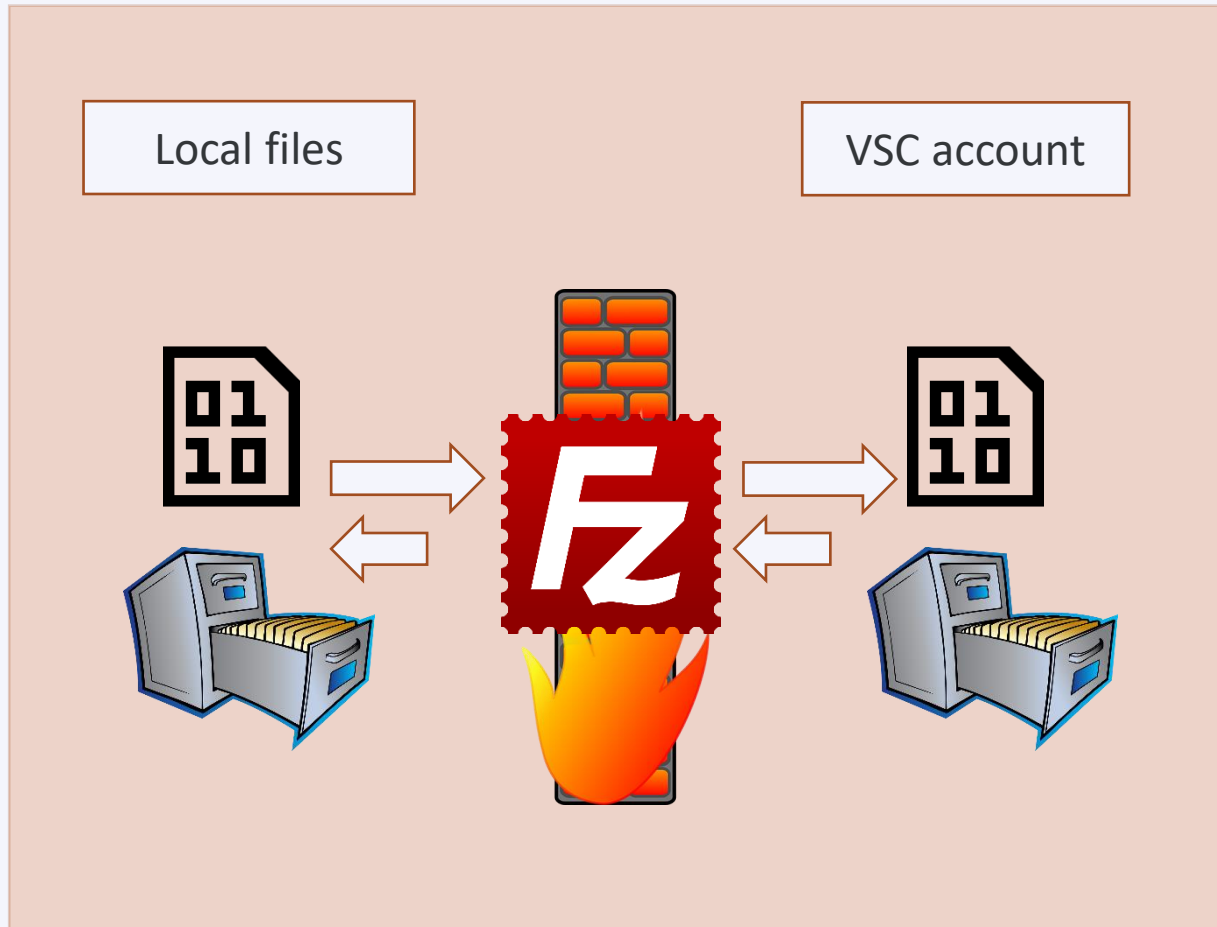
A terminal window showing the process of connecting to a VSC SSH certificate service. The user runs `ssh vsc32591@firewall.vscentrum.be`. The terminal displays a warning about the host's RSA key fingerprint, which the user accepts. It then shows the user logging in as `vsc32591` from `10.40.132.98:54800`. The connection is reset by the firewall. The user then runs `ssh vsc32591@login.hpc.kuleuven.be`. The terminal displays the `genius` logo and system information for the `login` role on a `ProLiant DL380 Gen10` server. It also shows the user's welcome message and the location of the documentation.





Data transfer

File transfer tools

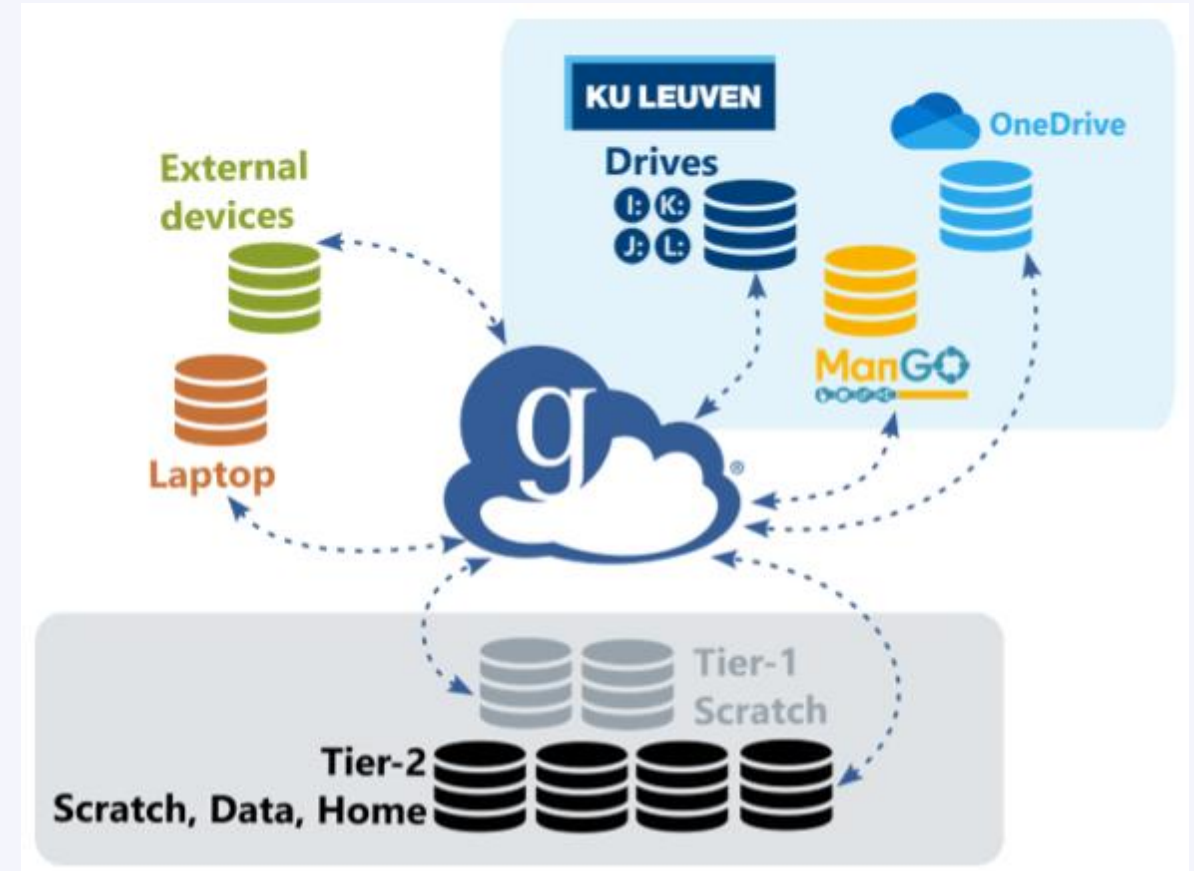


Application	OS
FileZilla	Windows, Linux, Mac
WinSCP	Windows
rsync, scp	Linux, Mac
Globus	Window, Linux, Mac

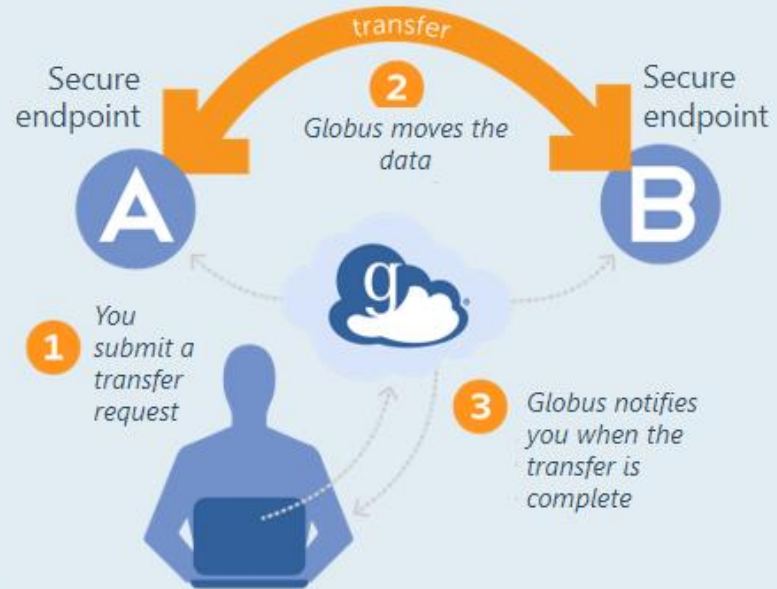
File transfer with Globus

- ✓ Web interface for transferring, sharing and searching within data
- ✓ Schedule and resume (large) transfers
- ✓ Workflow:
 - Define “endpoint” on the source
 - Define “endpoint” on the destination
 - Transfer between endpoints
- ✓ Endpoints on all VSC sites:
 - all Tier-2 clusters: home, data, scratch *
 - all Tier-1 clusters: scratch, projects

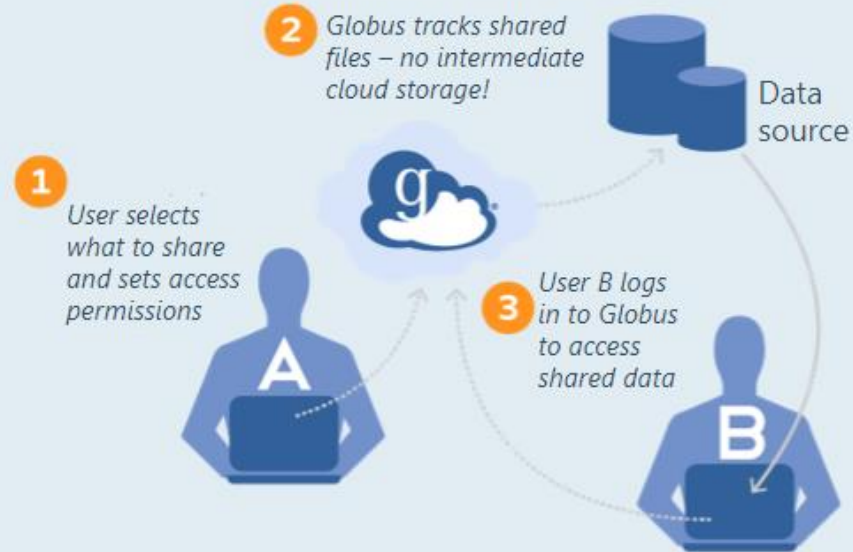
* You can also access project storage through this endpoint (Leuven Tier-2 only)



Transfer data



Share data





Open OnDemand

Open OnDemand

- ✓ Access clusters via your web browser
<https://ondemand.hpc.kuleuven.be>
- ✓ Authenticate via your institute's IDP
- ✓ File browser for home and data
- ✓ Job monitor
- ✓ Login shell access
- ✓ Interactive Apps

 Please close your interactive sessions when you no longer actively use them!

KU LEUVEN

Open OnDemand provides an integrated, single access point for all your HPC resources.

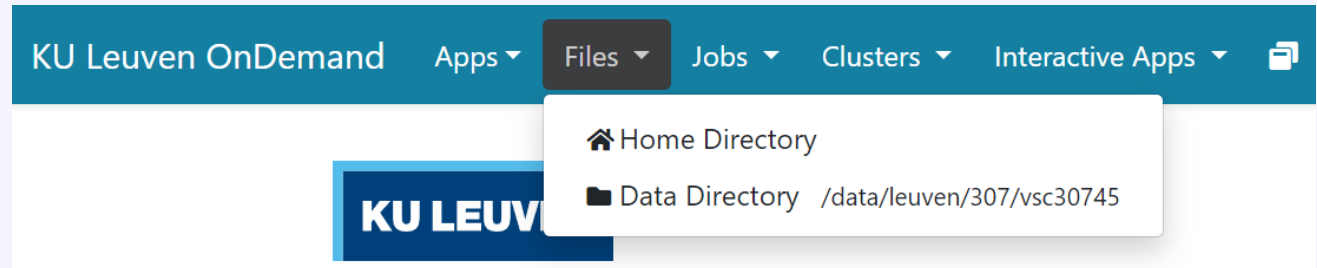
Pinned Apps A featured subset of **all available apps**

 Active Jobs System Installed App	 Home Directory System Installed App	 Login Server Shell Access System Installed App	 MATLAB System Installed App
 Fluent System Installed App	 Visual Studio Code System Installed App	 Desktop System Installed App	 gnuplot System Installed App
 IGV-Web System Installed App	 Interactive Shell System Installed App	 JupyterLab System Installed App	 ParaView System Installed App
 PyCharm System Installed App	 RStudio Server System Installed App	 TensorBoard System Installed App	 VS Code Tunnel System Installed App

Open OnDemand – File Browser

Access your VSC storage

- ✓ Use the “Files” tab to access your VSC_HOME and VSC_DATA
- ✓ VSC_SCRATCH is not available from this menu



File/Folder management

- ✓ OK for transferring small files/folders
- ✓ Files/folders open in the same browser tab
- ✓ Deleted files/folders can be retrieved from (hourly, daily, weekly) snapshots

Open OnDemand – Active Jobs

Monitor your active jobs

KU Leuven OnDemand Apps ▾ Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾   

🕒 Active Jobs
🔧 Project Manager

Your Jobs ▾ All Clusters ▾

Filter:

Active Jobs

Show entries

ID	Name	User	Account	Time Used	Queue	Status	Cluster	Actions
 62138076	sys/dashboard/sys/rstudio	vsc32591	lp_hpcinfo	00:00:04	interactive	Running	wice	

Showing 1 to 1 of 1 entries

Previous 1 Next

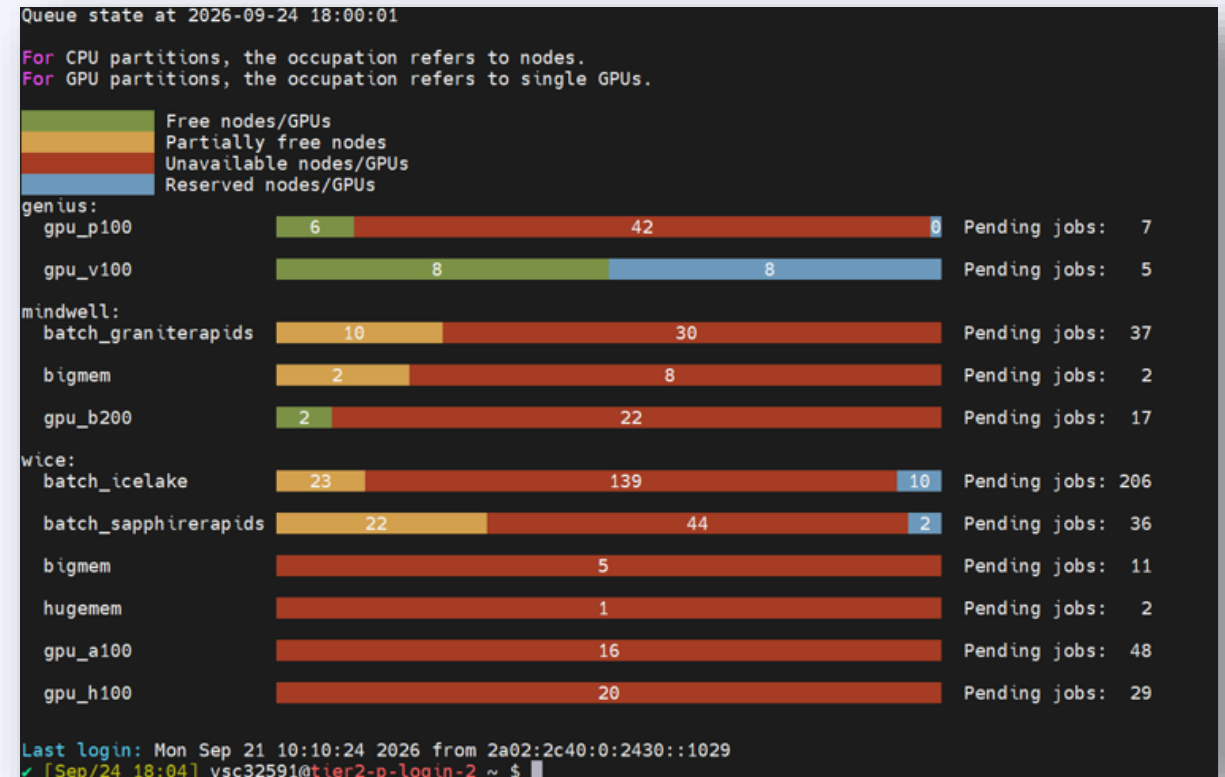


Open OnDemand – Login Server Shell Access

- ✓ Starts a shell in a new browser tab
- ✓ You land on a login node
- ✓ You land in your VSC_HOME
- ✓ Try:
\$ sam-balance
\$ myquota



Do NOT compute on login nodes





Open OnDemand – Interactive Shell

- ✓ Start a shell as a job (on a compute node)
- ✓ Recommended to:
 - Install your own software (in `VSC_DATA`)
 - Test/run your program interactively
 - Debug a program
- ✓ Choose the relevant cluster, partition and resources (cores, memory, GPU)

Interactive Apps

Servers

Interactive Shell

Jupyter Lab

Nvidia Rapids

RStudio Server

Tensorboard

code-server

Interactive Shell

This app will launch an interactive shell on a node.

Cluster

wice

Account

lpt2_sysadmin

Partition

interactive

batch(_long) or bigmem or interactive or gpu(_p100[_v100]) or dedicated...

Number of hours

1

Number of cores

1

Required memory per core in megabytes

3400

Number of nodes

1

Number of gpu's

0

[type:] Specify the total number of GPUs slices for the job. An optional GPU type specification can be supplied. For example "A100:3" or "3".

Reservation (optional)

Name of an existing reservation in which the job should run

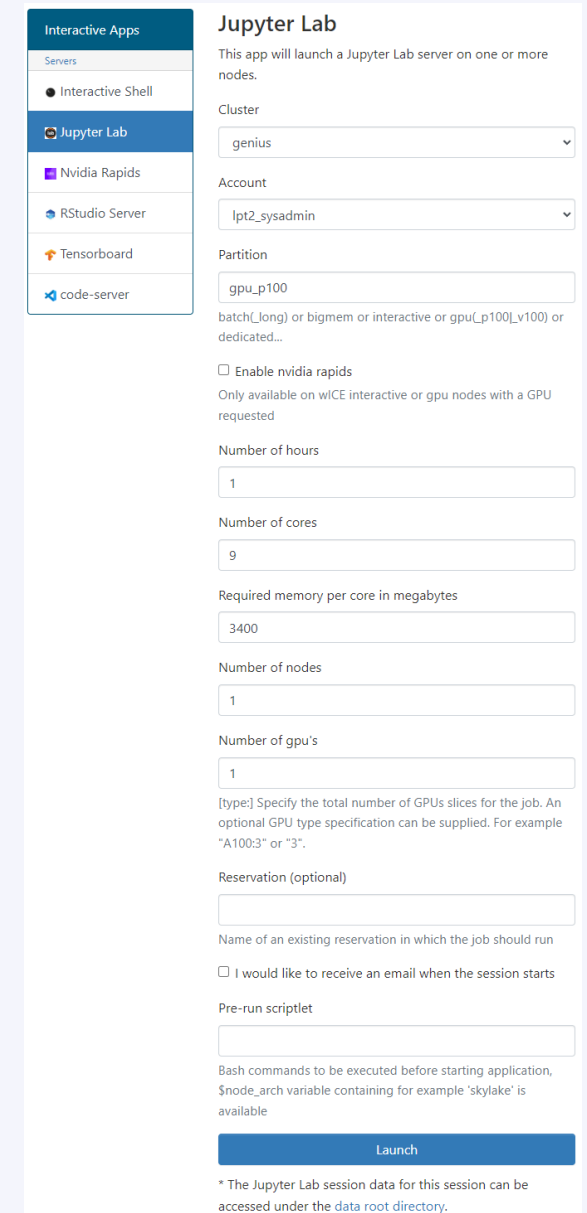
☐ I would like to receive an email when the session starts

Launch

* The Interactive Shell session data for this session can be accessed under the [data root directory](#).

Open OnDemand – JupyterLab

- ✓ Create a notebook as a job (on a compute node)
- ✓ Suitable for:
 - Pre/post-processing your data
 - Prototyping
 - Adding figures and text for pedagogical purposes
- ✓ Choose the appropriate cluster, partition and resources (cores, memory, GPU)
- ✓ User your own Python/R kernels (next slide)



The screenshot shows the 'Jupyter Lab' configuration page. On the left is a sidebar with 'Interactive Apps' and 'Servers' sections. 'Interactive Apps' includes 'Interactive Shell', 'Jupyter Lab' (selected), 'Nvidia Rapids', 'RStudio Server', 'Tensorboard', and 'code-server'. The main area is titled 'Jupyter Lab' and contains the following fields:

- Cluster:** A dropdown menu with 'genius' selected.
- Account:** A dropdown menu with 'lpt2_sysadmin' selected.
- Partition:** A text input field with 'gpu_p100' entered.
- batch[_long] or bigmem or interactive or gpu[_p100[_v100] or dedicated...:** A checkbox for 'Enable nvidia rapids' is unchecked. Below it is a note: 'Only available on wICE interactive or gpu nodes with a GPU requested'.
- Number of hours:** A text input field with '1' entered.
- Number of cores:** A text input field with '9' entered.
- Required memory per core in megabytes:** A text input field with '3400' entered.
- Number of nodes:** A text input field with '1' entered.
- Number of gpu's:** A text input field with '1' entered.
- [type:] Specify the total number of GPUs slices for the job. An optional GPU type specification can be supplied. For example "A100:3" or "3".:** A text input field is empty.
- Reservation (optional):** A text input field is empty.
- Name of an existing reservation in which the job should run:** A text input field is empty.
- I would like to receive an email when the session starts:** An unchecked checkbox.
- Pre-run scriptlet:** A text input field is empty.
- Bash commands to be executed before starting application, \$node_arch variable containing for example 'skylake' is available:** A text input field is empty.
- Launch:** A blue button.

At the bottom, a note states: '* The Jupyter Lab session data for this session can be accessed under the [data root directory](#).'



Open OnDemand – Custom R/Python kernels

- ✓ The default kernels may not offer the R/Python/... interpreter version or packages you want
- ✓ You can create a miniconda env and add it to your OnDemand kernels

Step 1: Miniconda

Start an Interactive Shell

Go to your data folder

```
$ cd ${VSC_DATA}
```

Download Miniconda

```
$ wget https://repo.continuum.io/miniconda/Miniconda3-latest-Linux-x86\_64.sh
```

Install Miniconda in your VSC_DATA

```
$ bash Miniconda3-latest-Linux-x86_64.sh -b -p ${VSC_DATA}/miniconda3
```

Permanently add the path to Miniconda to your ~/.bashrc

```
$ echo 'export PATH="${VSC_DATA}/miniconda3/bin:${PATH}"' >> ~/.bashrc
```



Open OnDemand – Custom R/Python kernels

Step 2: Environment Setup

Create a new environment `<env_name>` with all packages you need, e.g.

```
$ conda create --name=<env_name> numpy scipy ...
```

Enable your new environment

```
$ source activate <env_name>
```

Add the ipykernel package

```
$ conda install -c conda-forge ipykernel
```

Step 3: Register your kernel

```
$ python -m ipykernel install --user --env PYTHONPATH "" \
    --name <env_name> --display-name <env_name>
```

Step 4: Launch JupyterLab

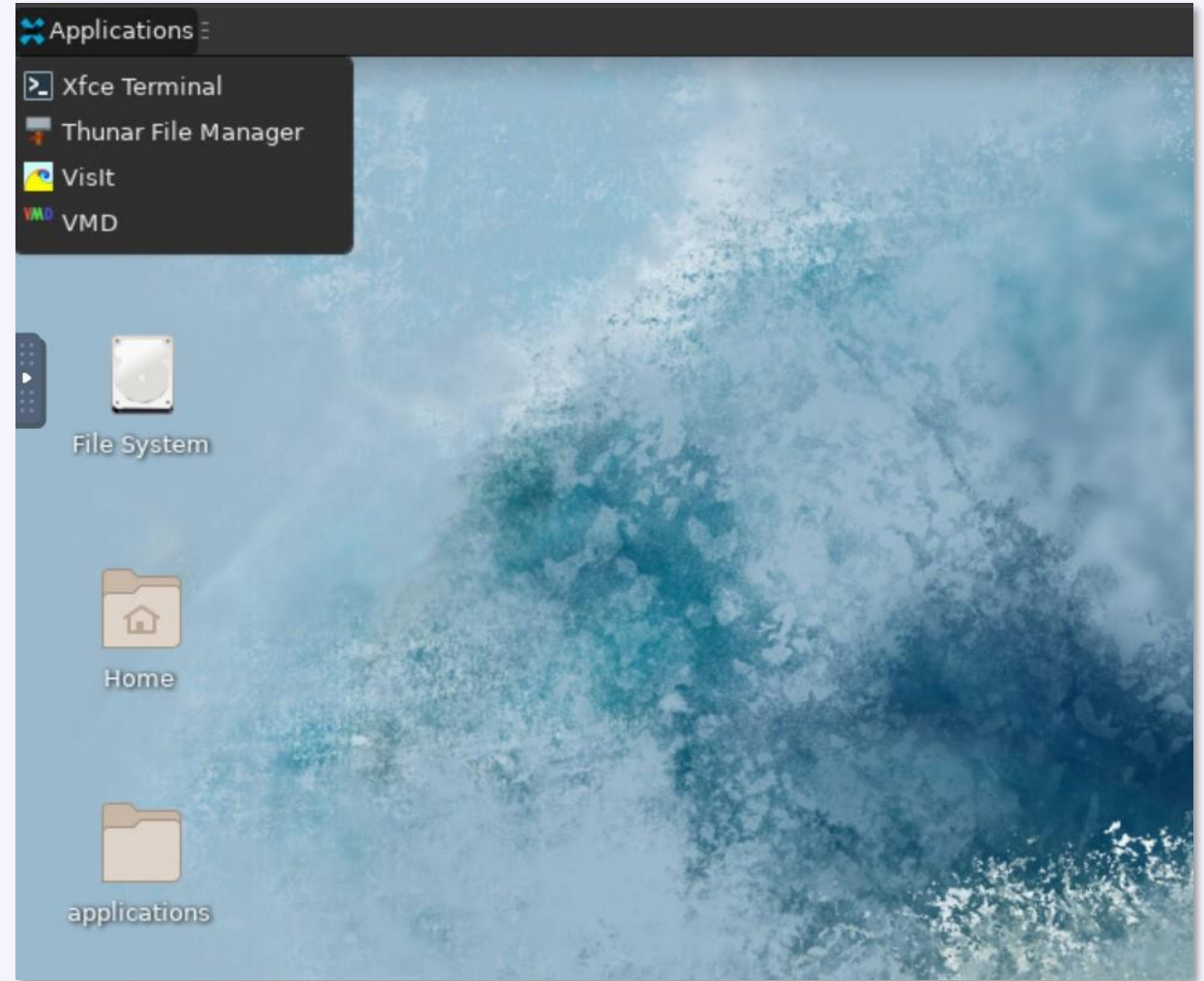
Now start a new Jupyter Lab session

Your new kernel should be available there



Open OnDemand – noVNC Desktop

- ✓ Ideal for apps with GUI
- ✓ Load (graphical) modules in Terminal, e.g. MATLAB, Fluent, ncview, VisIt, ...
- ✓ Minimalistic Terminal (different from login node)
- ✓ Two visualization apps:
 - VisIt
 - VMD





Open OnDemand – Other apps



Start Visual Studio Code server as a job

Develop, deploy and debug your workflow directly on HPC

Use of the interactive partitions is free of charge and is a good choice

Supports many languages + GitHub integration



Start RStudio IDE as a job

Install your own packages in VSC_DATA



Track machine learning metrics and visualize data during the workflow

You need to provide a log folder



Open OnDemand – Other apps



Start PyCharm IDE as a job

Develop, deploy and debug your Python workflow directly on HPC

Use Python environment based on Conda or virtual environments



Use Fluent for CFD simulations (single-node)

You need to be in a license group before being able to see the app logo

Contact your supervisor to help you joining an existing license group



Tunnel to the cluster from your local VSCode, so that you code locally, but run remotely! You can enable extensions such as Copilot and more.

Open OnDemand – Resources

All apps have (almost) similar resource form

E.g. to start a JupyterLab session:

- ✓ Pick a valid Slurm credit account
- ✓ Default partition for most apps: `interactive` (interactive: max 8 cores, 1 GPU instance, 16 hr)
- ✓ Using `interactive` partitions is free of charge

Interactive Apps

Servers

● Interactive Shell

 Jupyter Lab

 RStudio Server

 Tensorboard

 code-server

Work in progress

 ParaViewWeb - Work in progress

 cryosparc

Jupyter Lab

This app will launch a Jupyter Lab server on one or more nodes.

Account

lp_myproject

Partition

interactive

batch(_long) or bigmem or interactive or gpu or dedicated...

Number of hours

3

Number of cores

1

Required memory per core in megabytes

3400

Number of nodes

1

Number of gpu's

0

[type:] Specify the total number of GPUs slices for the job. An optional GPU type specification can be supplied. For example "A100:3" or "3".

Reservation (optional)

Name of an existing reservation in which the job should run

☐ I would like to receive an email when the session starts

Launch



Software

Software – Centrally installed modules

- ✓ OS: wICE is running on Rocky Linux 9.8
- ✓ Toolchains:
 - *intel* (icc, icpc, ifort; Intel MPI; Intel MKL)
 - *foss* (gcc, g++, gfortran; OpenMPI; ScaLAPACK, OpenBLAS, FFTW)
 - Starting toolchain-year on wICE: 2021a
 - Starting toolchain-year on Mindwell: 2024a



Not all toolchains are compatible with each other!

Never mix *foss* and *intel*

Command	Remark
<code>module av</code>	List all available modules
<code>module av Python</code>	List all Python-related modules
<code>module spider Python</code>	Get more info
<code>module load Python/3.12.3-GCCcore-13.3.0</code>	Load a specific module
<code>module list</code>	List all loaded modules and their dependencies
<code>module unload Python/3.12.3-GCCcore-13.3.0</code>	Unload a module and its dependencies
<code>module purge</code>	Remove all modules from your work session

Software – Your specific needs

- ✓ Use [existing modules](#)
- ✓ For missing module, [contact the HPC support team](#)
For Python and R packages, however, we prefer that you install it yourself
 - Read more about [Python Package Management](#)
 - Read more about [R Package Management](#)
- ✓ Lincensed software:
You are responsible for providing the license
Exceptionally, we provide MATLAB license (request joining `lli_matlab` group)
For license servers, please contact [SET-IT](#)
- ✓ If you want to compile software yourself:
 - Best install it in your `$VSC_DATA`
 - Guidelines on [VSCdocs](#)
 - Use either *intel* or *foss* toolchains
 - Compile your code on a compute node (with interactive job)
 - If you run into problems, don't hesitate to [contact us](#)



Starting to compute

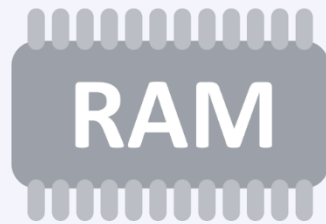
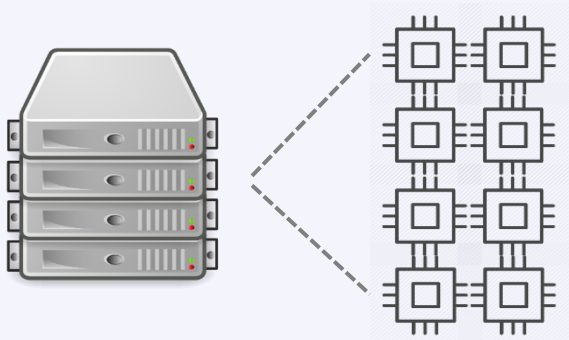
Resource glossary

- ✓ **Cluster:** which machine to use? [wICE](#) or [Mindwell](#)?
- ✓ **Nodes:** how many compute servers to allocate?
- ✓ **Cores:** how many cores per node to allocate?
- ✓ **Memory:** how much CPU memory to allocate?
- ✓ **Partition:** batch, gpu, bigmem, ...?
- ✓ **Walltime:** how long to allocate the resources?
- ✓ **Storage:** how much storage (data, scratch, etc) will the job require?
- ✓ **Credits:** how many compute credits will the job consume?



Default resources

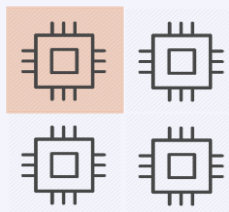
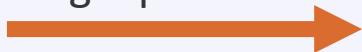
- ✓ **Cluster:** the cluster associated with the node you're on (wICE or Mindwell)
- ✓ **Nodes:** 1
- ✓ **Cores:** 1
- ✓ **Memory:** depends on which node(s) you use
- ✓ **Partition:** batch
- ✓ **Walltime:** 1 hour
- ✓ **Storage:** no default
- ✓ **Credits:** no default



Serial application

1 process on 1 core

Single process



One node

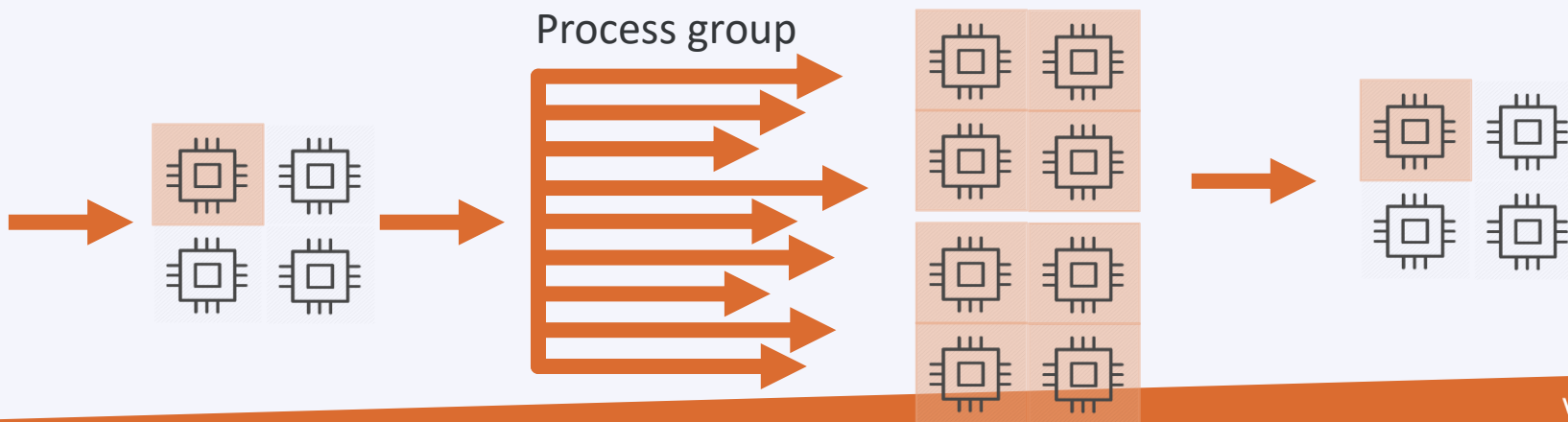
Multi-threaded application

N threads on N cores
on **1 node**

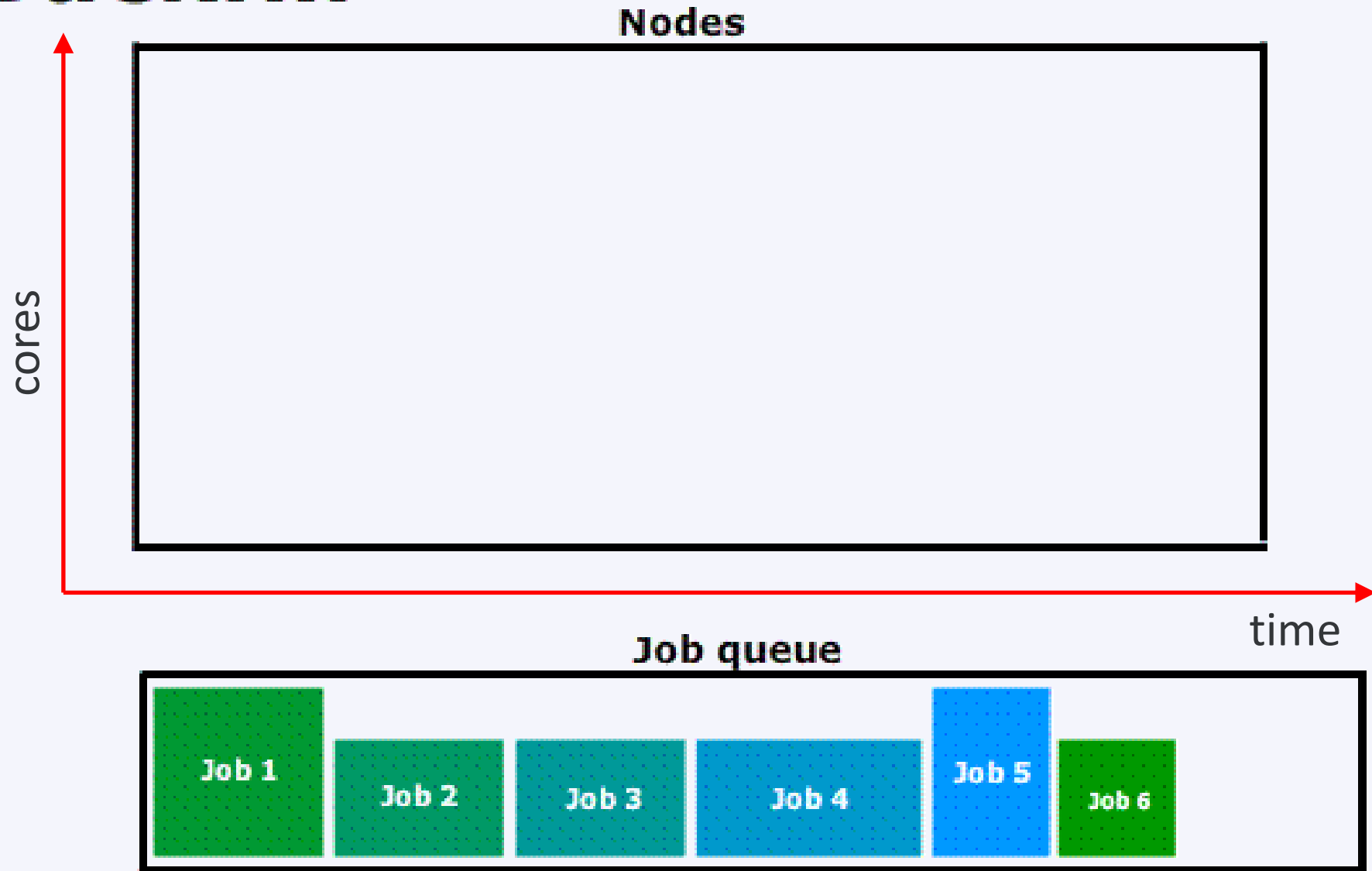


Distributed application

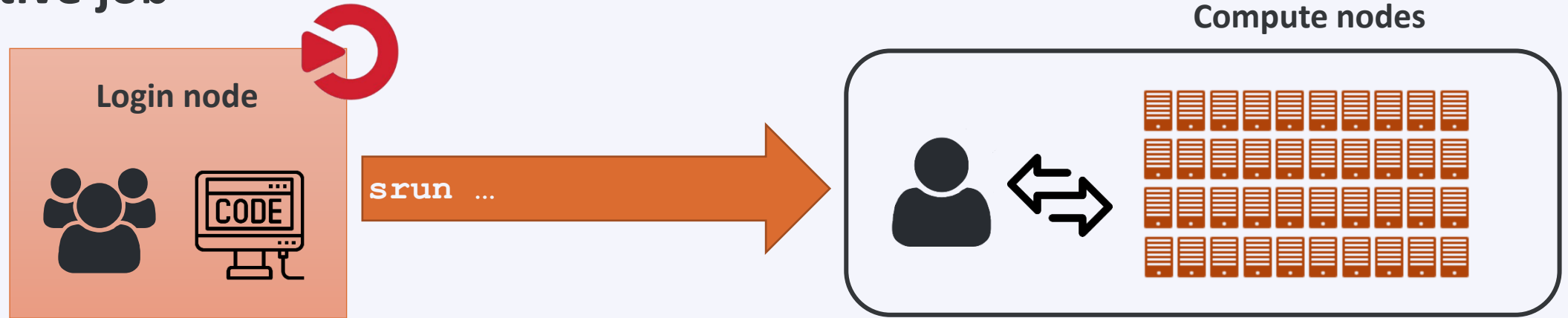
Many processes on
multiple cores,
possibly spread over
multiple nodes



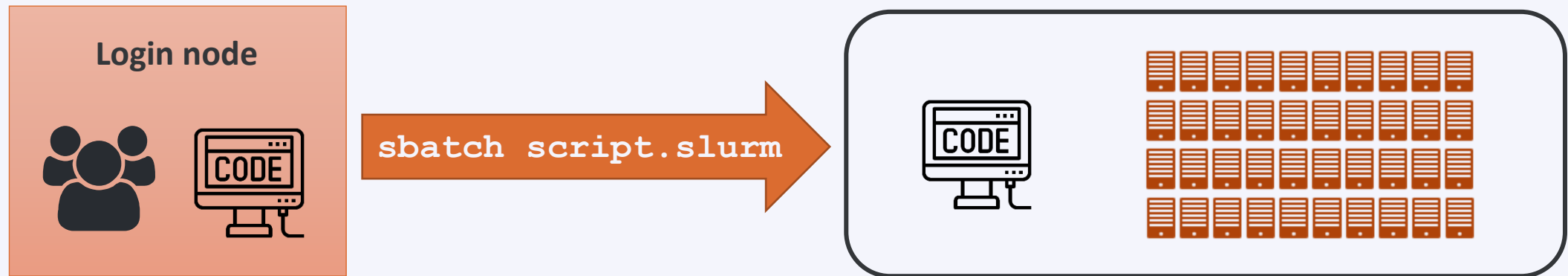
Backfill



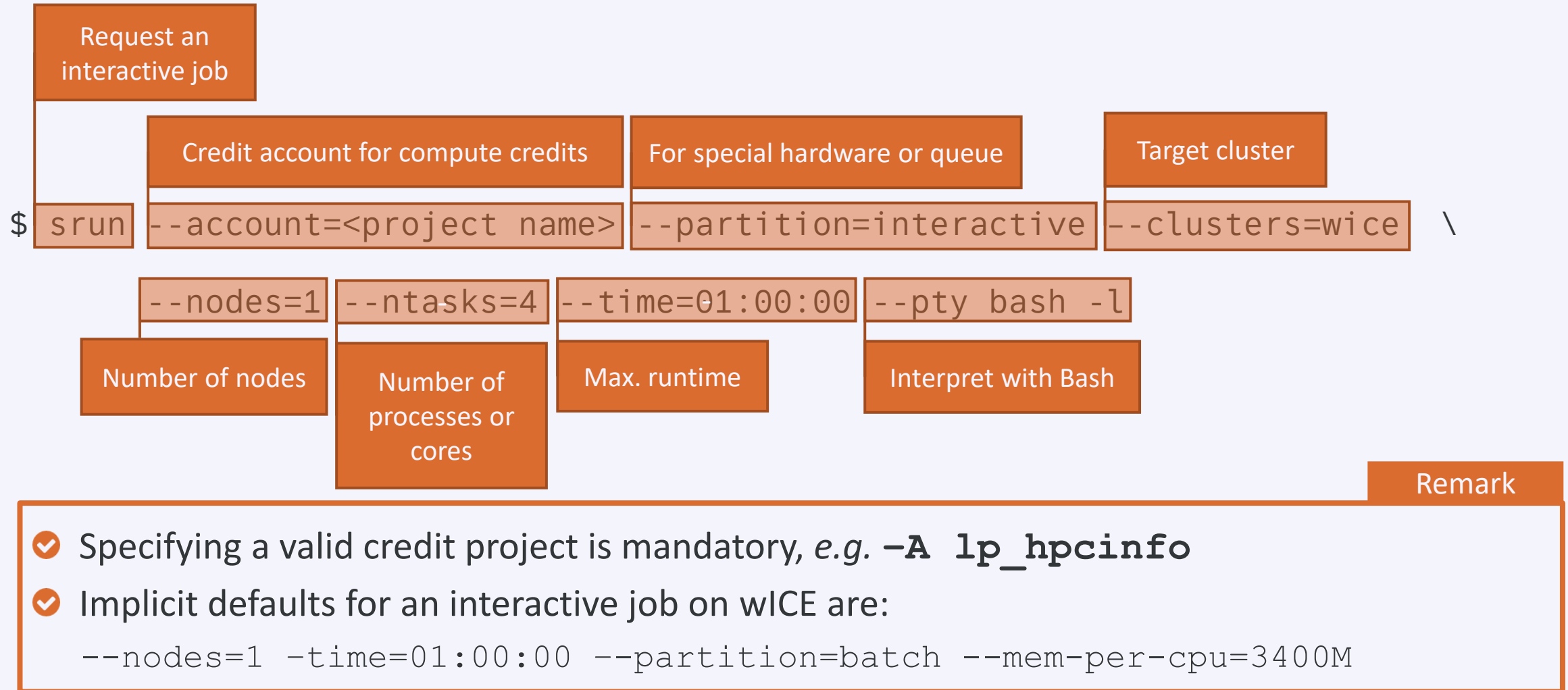
Interactive job



Batch job



Example interactive job on wICE



Slurm command-line tools

Command	Purpose
\$ sbatch ...	Submit a batch job
\$ srun ...	Submit an interactive job
\$ scancel --cluster=wice <JobID>	Cancel a specific pending or running job
\$ scontrol show job --clusters=wice <JobID> \$ slurm_jobinfo <JobID>	Detailed job info (very useful to diagnose issues)
\$ squeue --clusters=wice --me --long	Status of all recent jobs
\$ squeue --clusters=wice --me --start	Give a rough estimate of when your jobs will start
\$ sinfo --clusters=wice	Info about the state of available partitions and nodes
\$ sacct --clusters=wice --batch --job <JobID>	Show the jobscript used for a given batch job
\$ slurmtop \$ slurm_queuestate	Overview of the clusters
\$ sam-balance	Overview of all your available credit projects
\$ sam-list-allocations	Detailed overview of your credit allocation history

Slurm job submit options

Slurm option	Remarks
<code>--nodes=X --ntasks-per-node=Y</code> <code>--ntasks=X*Y</code> <code>--ntasks=X --cpus-per-task=Y</code>	or or (e.g. Hybrid MPI + OpenMP)
<code>--partition=<partition_name></code>	Default: "batch" partition
<code>--mem-per-cpu=<size><M></code>	Min memory per core, e.g. 3400M (wICE)
<code>--time=<dd-hh:mm:ss></code>	e.g. 1-12:30:00
<code>--job-name=<job_name></code>	
<code>--output=<file_template></code>	STDOUT; default="slurm-%j.out"
<code>--error=<file_template></code>	Default: redirect to STDOUT
<code>--mail-type=FAIL,BEGIN,END</code>	
<code>--mail-user=<email-address></code>	
<code>--export=<ALL, key=value></code>	Additional variables to pass to the job
<code>--account=<account_name></code>	Credit account (mandatory)

Slurm job submit options

Slurm option	Remarks
<code>--account=<account_name></code>	Mandatory (credits)
<code>--partition=<partition_name></code>	Default: "batch" partition
<code>--time=<dd-hh:mm:ss></code>	e.g. 1-12:30:00
<code>--nodes=X --ntasks-per-node=Y</code> <code>--ntasks=X*Y</code> <code>--ntasks=X --cpus-per-task=Y</code>	or or (e.g. Hybrid MPI + OpenMP)
<code>--mem-per-cpu=<size><M></code>	Min memory per core, e.g. 3400M (wICE)
<code>--job-name=<job_name></code>	
<code>--output=<file_template></code>	STDOUT+STDERR; default="slurm-%j.out"
<code>--mail-type=FAIL,BEGIN,END</code>	
<code>--mail-user=<email-address></code>	
<code>--export=<ALL,key=value></code>	Additional variables to pass to the job

Long and short options

Some (not all) of the `sbatch`, `srun` and `salloc` command line options also have a short version

Short version	Long version	Meaning
<code>-A</code>	<code>--account</code>	Slurm account name
<code>-a</code>	<code>--array</code>	Job array range
<code>-M</code>	<code>--clusters</code>	Cluster name
<code>-c</code>	<code>--cpus-per-task</code>	Number of cores per task (default: 1)
<code>-d</code>	<code>--dependency</code>	Job dependency (after, afterok, afterany, ...)
<code>-i</code>	<code>--input</code>	STDIN filename
<code>-e</code>	<code>--error</code>	STDERR filename
<code>-o</code>	<code>--output</code>	STDOUT filename
<code>-t</code>	<code>--time</code>	Maximum walltime
<code>-N</code>	<code>--nodes</code>	Minimum num nodes
<code>-n</code>	<code>--ntasks</code>	Maximum num tasks
<code>-p</code>	<code>--partition</code>	Partition name

Interactive jobs

- ✓ Interactive job with default options (1 core for 1 hour on the *batch* partition)

```
$ srun --clusters=wice --account=lp_hpcinfo --pty bash -l
```

- ✓ Interactive job with X-forwarding

```
$ srun --clusters=wice --account=lp_hpcinfo --x11 --pty bash -l
```

- ✓ Interactive job with multiple cores, on the *interactive* partition

```
$ srun --clusters=wice --account=lp_hpcinfo --nodes=1 --ntasks=4 \
    --partition=interactive --pty bash -l
```

- ✓ Interactive job with multiple cores and a GPU device

```
$ srun --clusters=wice --account=lp_hpcinfo --nodes=1 --ntasks=18 \
    --partition=gpu --gpus-per-node=1 --pty bash -l
```

- ✓ To join one of your running jobs (connecting to the first compute node in the job's allocation):

```
$ srun -M wice --jobid=<JobID> --overlap -pty bash -l
```

Example Slurm job script

jobscript.slurm

```
#!/bin/bash -l
#SBATCH --clusters=wise
#SBATCH --account=lp_hpcinfo
#SBATCH --nodes=1
#SBATCH --ntasks=4
#SBATCH --mem-per-cpu=3400M
#SBATCH --job-name=hpc_workflow
#SBATCH --mail-type=BEGIN,END,FAIL
#SBATCH --mail-user=my.name@kuleuven.be
```

Shebang

Resource list

```
module load intel/2024a
module load Python/3.12.3-GCCcore-13.3.0
which python
cd $VSC_SCRATCH/projects/simulations
cp -r $VSC_DATA/input_data .
```

Loading modules

Move data

```
python modelling.py
```

Execute commands

```
cp -r output_data $VSC_DATA
rm -rf ./input_data ./output_data
```

Move data

Batch job workflow

- ✓ Submit the job to the scheduler
- ✓ Receive a unique JobID
- ✓ Error and output files

Job script

```
#!/bin/bash -l  
...
```

Submit command

```
$ sbatch simulation.slurm
```

JobID

```
Submitted batch job 60042478 on  
cluster wice
```

stderr, stdout

```
$ ls *.out  
slurm-60042478.out
```

Output files

- ✓ STDERR and STDOUT are by default redirected to a single file: `slurm-<JobID>.out`
- ✓ Contains job info, all errors and warnings, and printouts
- ✓ Worth going through (especially if it's a new kind of job)
- ✓ Address all warnings and errors (if you can)
- ✓ Typical error examples on the right

STDOUT + STDERR

```
$ ls slurm-*.out  
slurm-60042478.out
```

Out of memory

```
slurmstepd: error: Detected 1 oom-kill event(s).  
Some of your processes may have been killed by the  
cgroup out-of-memory handler.
```

Short walltime

```
slurmstepd: error: *** JOB 60042478 ON s28c11n2  
CANCELLED AT 2023-02-08T10:03:43 DUE TO TIME LIMIT  
***
```

Low disk space

```
IOError: [Errno 122] Disk quota exceeded
```

Output files

- ✓ Standard output and error channels can be redirected to other files:

```
#SBATCH --output ...
```

```
#SBATCH --error ...
```

stdout

```
$ ls slurm-*.out  
slurm-60041238.out
```

Output file

```
SLURM_JOB_ID: 60033947  
SLURM_JOB_USER: vscXXXXX  
SLURM_JOB_ACCOUNT: lp_wice_pilot  
SLURM_JOB_NAME: testjob  
SLURM_CLUSTER_NAME: wice  
SLURM_JOB_PARTITION: batch  
SLURM_NNODES: 1  
SLURM_NODELIST: m28c30n4  
SLURM_JOB_CPUS_PER_NODE: 72  
Date: Tue Jan 10 17:02:04 CET 2023  
Walltime: 00-01:00:00  
=====  
/apps/leuven/rocky8/icelake/2021a/  
softwar  
e/intel-compilers/2021.2.0/compile  
r/2021.2.0/linux/bin/intel64/icc  
cp: cannot stat '/apps/leuven/trai  
ning/test': No such file or directo  
ry  
Hello World
```

Resource
summary

stdout

Job monitoring

Example

```
$ scontrol show job --clusters=wice 60049330
  JobId=60049330 JobName=f70.slurm
  UserId=vsc3...(253...) GroupId=vsc3...(253...)
Account=lp_my_project QOS=lp_my_project
JobState=PENDING Reason=QOSGrpBillingMinutes
  SubmitTime=2023-02-16T00:24:58 EligibleTime=2023-02-16T00:24:58
  Partition=batch NodeList= NumNodes=4-4 NumCPUs=288 NumTasks=288 CPUs/Task=1
  TRES=cpu=288,mem=979200M,node=4,billing=733
  MinCPUsNode=72 MinMemoryCPU=3400M
  WorkDir=/vsc-hard-mounts/leuven-data/3../vsc3../Odonate_SOM
  StdErr=/vsc-hard-mounts/leuven-data/3../vsc3../Odonate_SOM/slurm-60049330.out
  StdIn=/dev/null
  StdOut=/vsc-hard-mounts/leuven-data/3../vsc3../Odonate_SOM/slurm-60049330.out
```

Diagnosis

Why does my job remain in a pending state?

Check out the “Reason” on Slurm docs: https://slurm.schedmd.com/resource_limits.html

Other wICE partitions

GPU

```
#SBATCH --partition=gpu
#SBATCH --nodes=1
#SBATCH --ntasks=18
#SBATCH --gpus-per-node=1
```

Big memory

```
#SBATCH --partition=bigmem
#SBATCH --nodes=1
#SBATCH --tasks-per-node=72
#SBATCH --mem-per-cpu=28000M
```

Interactive partitions

- ✓ Accessible via command line and Open OnDemand
- ✓ Short queue time
- ✓ Max resource per user: 8 cores, 1 GPU instance, 16 hour walltime
- ✓ Free of charge
- ✓ Hardware:
 - wICE: 4 nodes with 64 cores/node, 512 GiB mem, 1 physical GPU (split in 7 instances)
 - Mindwell: 2 nodes with 192 cores/node, 768 GiB mem, 2 Physical GPUs
- ✓ Purpose:
 - To quickly compile, test, debug your (parallel) application
 - To pre- or postprocess your data (incl. visualizations)

```
$ srun --account=lp_myproject --clusters=wice \  
    --partition=interactive --nodes=1 --time=16:00:00 \  
    --gpus-per-node=1 --pty bash -l
```

Debug partitions

- ✓ Accessible via command line and Open OnDemand
- ✓ Quickly test if your (parallel) application works
- ✓ Short queue time
- ✓ Only one job at a time, max waltime: 1 hr

Cluster	Partition	Resources
wICE	interactive	4 Icelake nodes, each with 7 MIG instances of 1 A100 GPU
	gpu_p100_debug	1 Skylake node with 4 P100 GPU
	gpu_a100_debug	1 Icelake node with 1 A100 GPU
Mindwell	interactive	2 Granite Rapids nodes, each with 2 RTX5000 GPUs

Containers: Apptainer (Singularity)

✓ What?

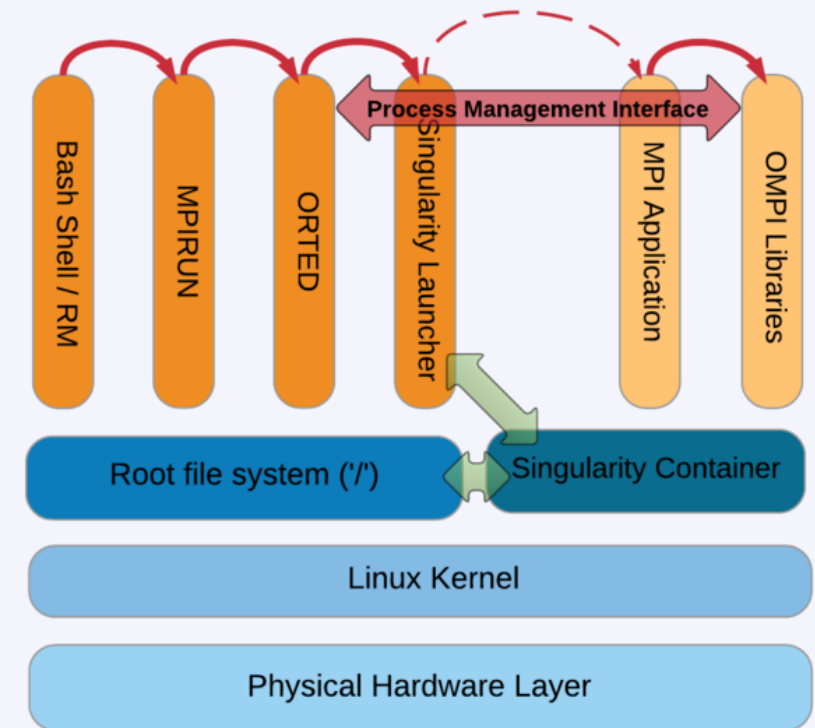
- Self-contained OS image in which you can install software and include data

✓ Why?

- Package managers like apt can be convenient
- Portability: images can be used on various hardware (but performance may be sub-par)

✓ How?

- Create the image
- Run it on wICE or Mindwell
- Parallelization should work (multi-node MPI runs may require some care)



script.slurm

```
#!/bin/bash -l
#SBATCH --time=30:00
#SBATCH --nodes=1
#SBATCH --ntasks=72
#SBATCH --clusters=wice
#SBATCH --account=lp_hpcinfo
apptainer run Project.sif ./model.exe
```

High-throughput computing with *atools*

- ✓ You can use atools to handle high-throughput computing workloads
- ✓ Simply load the module:
`module load atools/<version>`
- ✓ VSC training: Workflows for Data Science
- ✓ Training material: <https://github.com/gjbex/Workflows-for-HPC>
- ✓ Recording: <https://www.youtube.com/watch?v=GlwaRyi1LVA>
- ✓ User documentation: <https://atools.readthedocs.io/en/latest>



Demo: try things yourself

Demo: try it yourself

- ✓ Transfer a file via the Open OnDemand Filebrowser
- ✓ Check your disk quota
- ✓ Check your available credits
- ✓ Check/load/list/unload/purge module

Demo: try it yourself

- ✓ Copy `/apps/leuven/training/HPC_intro/` to your `$VSC_DATA`
- ✓ Submit `cpujob.slurm` to the cluster
- ✓ List all your jobs with `squeue -M all`
- ✓ Check the information about the job with `slurm_jobinfo -M wice <job_ID>`
- ✓ Modify the `mpi.slurm` script to request 1 node, 72 cores for 30 minutes, with job start/end e-mail notifications
- ✓ Check the status of all the jobs

Demo: job monitoring

✓ Submit an interactive job

- Run your program on a compute node
- Open a new terminal and `ssh` to the allocated compute node
- Check the resources usage with `htop`

✓ Submit a GPU batch job

- While the job is running, check its details with `slurm_jobinfo`
- and check resource on the compute node using `ssh`, `htop` and `nvidia-smi`

Useful links

- ✓ Helpdesk (email): hpcinfo@kuleuven.be
- ✓ Helpdesk (online form): https://admin.kuleuven.be/icts/HPInfo_form/HPC-info-formulier
- ✓ Contact VSC (Tier-1, compute, data, cloud): https://docs.vscentrum.be/contact_vsc.html
- ✓ VSC website: <https://www.vscentrum.be>
- ✓ VSC documentation: <https://docs.vscentrum.be>
- ✓ VSC agenda, training sessions, events (User Day): <https://www.vscentrum.be/vsctraining>
- ✓ System status page: <https://status.vscentrum.be>

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